L03 — study notes

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Lecture 3 — Dynamic games: backward induction and subgame perfection · Map of the notes

Estimated unit. Built on 2026-10-16 from the study guide before the lecture. Reconcile with the slides when Aula/Lecture03.pdf arrives (/notes 03 –reconcile).

One claim. When a player can see earlier moves, a strategy becomes a complete contingent plan, and a game gains nodes that equilibrium play never visits. **Nash equilibrium does not check what players would do at those nodes**, so it admits threats nobody would carry out. **Subgame perfection requires a Nash equilibrium in every subgame**, reached or not, which under perfect information is just backward induction. **The economics is credibility**: a threat counts only if carrying it out is optimal when the moment comes, and **a player can gain by committing**, by moving first (Stackelberg) or by making one of its own payoffs worse (a bond).

Three layers per topic:

Layer	What it means here
U — intuition	the sequential Battle of the Sexes story, Gibbons's "noncredible threats", the Centipede evidence, Cabral's entry and commitment stories
C — course (examinable)	Tadelis Definitions 7.1–7.8 and 8.1–8.3, Proposition 8.1, the examples of §8.3 worked in full; Gibbons's outcome vs equilibrium; Cabral §7.2's numbers
G — formal	MWG §7.C (formal tree, perfect recall), §9.B (Definitions 9.B.1–9.B.2, Propositions 9.B.1–9.B.3). **Named, not examinable** unless the lecture brings them

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Reading order

Part I — describing a dynamic game (Tadelis Ch. 7)

1. L03-01-extensive-form — §7.1, pp. 129–137: the seven elements; Definitions 7.1–7.3; trust game, sequential vs simultaneous BoS, card game. Settles **"sequential" means a later mover *sees* the earlier move**, and **every simultaneous game has imperfect information**.
2. L03-02-strategies — §7.2.1–7.2.2, pp. 137–142: Definitions 7.4–7.7; $|S_i| = \prod m_k$; mixed vs behavioral with Fig. 7.9's numbers; Kuhn; the absent-minded driver. Settles **a strategy names an action at every information set, even unreachable ones**.
3. L03-03-normal-form-and-paths — §7.2.3–7.3, pp. 143–146; Gibbons pp. 116–128: the 2×4 matrix, the three Nash equilibria (O, oo), (O, of), (F, ff), Definition 7.8, and the **threat at an unreached node** that holds (F, ff) up.

Part II — the refinement (Tadelis Ch. 8.1–8.2)

4. L03-04-sequential-rationality-backward-induction — §8.1, pp. 151–153; §8.3.1, pp. 159–160; Gibbons §2.1.A: Definition 8.1, **Zermelo / Proposition 8.1**, the BoS and Gibbons games solved step by step, **the Centipede unravelling to** $(1, 1)$.
5. L03-05-subgame-perfection — §8.2, pp. 153–158; §8.3.3, pp. 163–166; Gibbons §2.2.A, §2.4.B: Definitions 8.2–8.3, which nodes start subgames, the **replace-and-repeat method**, voluntary BoS ($E_{\text{SPE}} = \{(YO, o), (NF, f)\}$), **MAD's two SPE**, and **outcome vs equilibrium**.

Part III — applications

6. L03-06-credibility-entry-commitment — Cabral §7.2, pp. 167–171; MWG Example 9.B.1: the entry game, the non-credible price war, **a bond worth 30**, the threshold $B^* = D_I - F_I$, commitment as moving first.
7. L03-07-stackelberg — §8.3.2, pp. 160–163; Gibbons §2.1.B: every step of (8.1)–(8.2), **(45, 22.5) vs (30, 30)**, why the leader must gain, the Nash-but-not-SPE profiles, and **why moving first is worth nothing unobserved**.
8. L03-08-time-inconsistency — §8.3.4, pp. 166–169: exponential vs hyperbolic discounting, the game against one's future self, and **a correction to the printed optimum** ($x_1 = K/2$, not $2K/3$).

Support

9. L03-09-formulas — the specification mapped item by item, every definition and every number.
10. L03-10-pitfalls — ten confusions with counterexamples and 26 original True/False items.
11. L03-11-python — a game-tree solver (strategies, Nash, subgames, SPE, backward induction), Stackelberg and the hyperbolic example, all with asserts against the books.

The one result to hold

SPE $\subseteq$ NE,	and in finite games of perfect information	SPE = backward induction

Every SPE is a Nash equilibrium; a Nash equilibrium is subgame perfect only if its plans are optimal at every node, including the nodes it never reaches. Under perfect information that test is backward induction (Fact, p. 158); otherwise solve the last subgames, replace them by their equilibrium payoffs, and repeat, branching when a subgame has several equilibria.

Numbers and cases to have memorized

Case	Nash equilibria	Subgame perfect	Page
Sequential BoS	$(O, oo), (O, of), (F, ff)$	(O, of)	145, 157
Voluntary BoS	$(YO, o), (NO, f), (NF, f)$	$(YO, o), (NF, f)$	158
MAD	six pure	$(IR, Nr), (ED, Bd)$	164, 166
Centipede	—	outcome $(1, 1)$, not $(3, 3)$	159
Cabral entry	$(e, \bar{r}), (\bar{e}, r)$	$(e, \bar{r})$; with a bond of 40: out, value **30**	168–170

Case	Nash equilibria	Subgame perfect	Page
Stackelberg, $p = 100 - q$, $c = 10$	a continuum (every point of $q_2 = \frac{90-q_1}{2}$ is a Nash outcome)	$(45, \frac{90-q_1}{2})$: profits **1012.5, 506.25** vs Cournot **900, 900**	161–163

Figures

Built by `figures/make_figures.py` (run: `python Notas/L03/figures/make_figures.py`; ends with an `assert` per figure).

Figure	Shows	Lives in
L03-bos-sequential-vs-simultaneous	the same tree with two information structures; $\|S_2\|$ 4 vs 2	L03-01-extensive-form §4
L03-bos-normal-form	the 2×4 matrix, three NE boxed, the SPE shaded	L03-03-normal-form-and-paths §2
L03-centipede-backward-induction	stop at every node, step by step	L03-04-sequential-rationality-backward-induction §4
L03-voluntary-bos-subgames	the two proper subgames and the two reduced games	L03-05-subgame-perfection §4.1
L03-entry-commitment	Cabral Figs. 7.7–7.8: the non-credible threat and the bond	L03-06-credibility-entry-commitment §3
L03-stackelberg-vs-cournot	reaction curves, isoprofits, the leader's reduced profit	L03-07-stackelberg §2
L03-hyperbolic-discounting	self 1's objective and four consumption plans	L03-08-time-inconsistency §2.2

Interactive companions

- The First-Mover Desk (`companions/companion-stackelberg-vs-cournot.html`) — a, b, c_1, c_2 move the reaction curves; read-outs for leader and follower quantities, prices and profits in both games, with the **first-mover advantage** as the answer (+112.5 at Tadelis's numbers) and the follower's shut-out thresholds as landmarks. Linked from L03-07-stackelberg.
- The Credible Threat Desk (`companions/companion-entry-commitment.html`) — the six payoffs of Cabral's entry game with a bond; the tree recolours with the SPE, the **value of commitment** is the answer (30 at Cabral's numbers), and a second panel shows the jump at B^*. Linked from L03-06-credibility-entry-commitment.

Both are checked by `companions/check_companions.py` (Python vs the pages' own JavaScript under Node).

Scope and ceiling

- **Examinable (C):** Tadelis Ch. 7–8 definitions, Proposition 8.1, the §8.3 examples; Gibbons's outcome/equilibrium distinction; Cabral §7.2's entry and commitment argument; Stackelberg vs Cournot with linear demand.

- **Named only (G):** MWG's formal tree and perfect recall (§7.C), Propositions 9.B.2–9.B.3, the absent-minded driver.
- **Next lectures, not here:** multistage games with simultaneous stages in depth and bargaining (Lecture 4), infinite horizons and repeated games (Lecture 5), beliefs at information sets / perfect Bayesian equilibrium (Lecture 8).
- **Out** (CLAUDE.md scope block): mechanism design, cheap talk, moral hazard; Tadelis exercise statements are cited by number and page only.

Course-wide position

Lecture	Relation to this one
Lecture 1	dominance and IESDS: backward induction is iterated deletion of non-optimal moves, node by node
Lecture 2	Nash equilibrium and mixed strategies: the solution concept refined here, and the equilibria of each subgame
Lecture 3	**extensive form, backward induction, subgame perfection, credibility, Stackelberg**
Lecture 4	multistage games and bargaining: Rubinstein solved by subgame perfection in an infinite horizon
Lecture 5	repeated games: finite repetition unravels by backward induction; infinite repetition sustains cooperation as an SPE
Lecture 6	collusion and relational contracts: credible punishments, commitment without courts
Lecture 7	Bayesian games: Nature's move and information sets become types
Lecture 8	signaling and PBE: subgame perfection is too weak when information sets are not singletons; beliefs are added

Rules pack: rules/game-theory (items 1, 2, 3, 7).

Bibliography actually used in these notes

Book	Where
Tadelis (2013) Ch. 7–8, pp. 129–174 — the spine (slides follow Tadelis)	everywhere
Gibbons (1992) §2.1.A–C (pp. 57–68), §2.2.A (pp. 71–73), §2.4 (pp. 115–129)	U layer of notes 03–05, 07
Cabral (2017) §7.2 (pp. 167–171)	note 06
MWG §7.C (pp. 221–225), §9.B (pp. 268–281)	G layer of notes 01, 04–06

Book	Where
Muñoz-Garcia & Toro-González (2019) Ch. 4 (pp. 153–214)	exercise pointers

Related project material

- Listas/pratica-aula-01-02.md — items on Cabral §7.2 (pp. 167–174), Cabral 7.14 Centipede (p. 182), MWG 9.B.13 (bargaining, pp. 302–303).
- Map/exercises-index.md and Map/study-guide.md — the Lecture 3 block and its after-class list (Muñoz-Garcia Ch. 4; Tadelis Ch. 7–8 exercises).
- Exercises by number and page: Tadelis 7.1–7.10 (pp. 147–150), 8.1–8.16 (pp. 170–174); Muñoz-Garcia Ch. 4 Exercises 1–20 (pp. 154–207; 16–17 are bargaining, for Lecture 4; 21 is moral hazard, out of scope).
- No quizzes, videos or slides for Lecture 3 exist yet.

7.1 · The extensive form: who moves when, and who knows what

> **Claim.** A normal form lists strategies and payoffs; it cannot say *when* a player moves or *what she has seen*. The extensive form adds both, and the second is the one that matters: **a game is sequential because a later mover observes an earlier move, not because the moves happen at different clock times.** Information sets are the device that records what a player knows. Prev: — · Next: L03-02-strategies · Map: L03-00-map

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1 · U — why the normal form is not enough

Tadelis opens Ch. 7 (p. 129) with the Battle of the Sexes from Lecture 2, changed in one detail:

	Chris: O	Chris: F
Alex: O	$2, 1$	$0, 0$
Alex: F	$0, 0$	$1, 2$

Alex finishes work at 3 p.m., goes to a venue and phones Chris at 4:45: "I am here." Chris now **knows** where Alex is.

1. Chris's payoff from joining Alex is 1 (if Alex is at the opera) or 2 (at the football); from going elsewhere, 0. So a rational Chris goes where Alex is, whatever that is.
2. Alex knows this (common knowledge of rationality). Going to the opera gives Alex 2; going to the football gives Alex 1. Alex goes to the opera.

The simultaneous game had two pure equilibria and a mixed one, with no way to choose among them. The sequential version has **one prediction, (O, o), the one Alex likes**: a *first-mover advantage*.

> **Moving first is not always an advantage.** In Matching Pennies the player who shows the coin first loses for sure (p. 130). What decides is the structure of payoffs, and the extensive form is what lets us compute it.

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2 · C — the seven elements of an extensive-form game (§7.1, pp. 130–131)

#	Element	Already in the normal form?
1	Set of players N	yes
2	Payoffs over outcomes $\{v_i(\cdot)\}_{i \in N}$	yes
3	**Order of moves**	no
4	**Actions available when a player moves**	partly (the action sets)
5	**What a player knows when she moves**	no
6	Probability distributions over exogenous events (**moves of Nature**)	no
7	The structure 1–6 is **common knowledge**	yes (complete information)

Gibbons states the same list in four items: players; when each moves, what each can do, what each knows; payoffs (Definition, pp. 115–116).

> **Element 5 is the one that separates "Chris moves after Alex and knows what Alex did" from "Chris moves after Alex but does not know what Alex did" (p. 131).** The second is the simultaneous game, whatever the clock says.

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3 · C — game trees (Definition 7.1, p. 133)

> **Definition 7.1.** A game tree is a set of nodes $x \in X$ with a precedence relation $x > x'$ ("x precedes x'"). Every node has only one predecessor. The relation is transitive, asymmetric and incomplete. The **root** x_0 precedes every other node. Nodes that precede no other node are **terminal nodes**, the set $Z \subset X$; payoffs $v_i : Z \to \mathbb{R}$ are attached to them. Every non-terminal node is assigned either to a player, $i(x)$, with action set $A_i(x)$, or to Nature.

What each clause buys:

Clause	Consequence
one predecessor per node	**a unique path from the root to every terminal node**, so an outcome *is* a path (used in §7.3, L03-03-normal-form-and-paths)
transitive, asymmetric	the tree has no cycles: play only moves forward
incomplete	two nodes on different branches are not ordered: neither comes "before" the other
payoffs on Z only	payoffs depend on the whole path, not on single moves

Worked: the trust game (Fig. 7.1, p. 132). Player 1 trusts (T) or not (N). N ends the game at $(0, 0)$. After T, player 2 cooperates (C): $(1, 1)$, or defects (D): $(-1, 2)$.

Terminal node	Path	v_1	v_2
z_1	N	0	0
z_2	T, C	1	1
z_3	T, D	-1	2

Solve it forward from what player 2 would do: after T, D pays player 2 a 2 against 1 from C, so player 2 defects; anticipating -1, player 1 picks N and gets 0. **Both would prefer (T, C) with $(1, 1)$; it is not reached.** This is the logic of L03-04-sequential-rationality-backward-induction already at work.

Tadelis's drawing convention, followed in every figure here: play flows top to bottom, and payoffs are a vertical list, player 1 on top.

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4 · C — information sets (Definition 7.2, p. 135)

> **Definition 7.2.** Every player i has a collection of information sets $h_i \in H_i$ that partition the nodes at which i moves, with: 1. if $h_i = \{x\}$ is a singleton, i knows she is at x; 2. if

$x \neq x'$ and both are in h_i, i does not know whether she is at x or x'; 3. if $x \neq x'$ and both are in h_i, then $A_i(x) = A_i(x')$.

Property 3 is a consistency requirement, not an assumption about the world. If two nodes offered different actions, the player could tell them apart by looking at her menu, and they could not be in one information set (p. 135; Gibbons p. 119 makes the same point).

Information sets are drawn either as an ellipse around the nodes or as a dashed line joining them (Fig. 7.4, p. 135). The figures here use the dashed line.

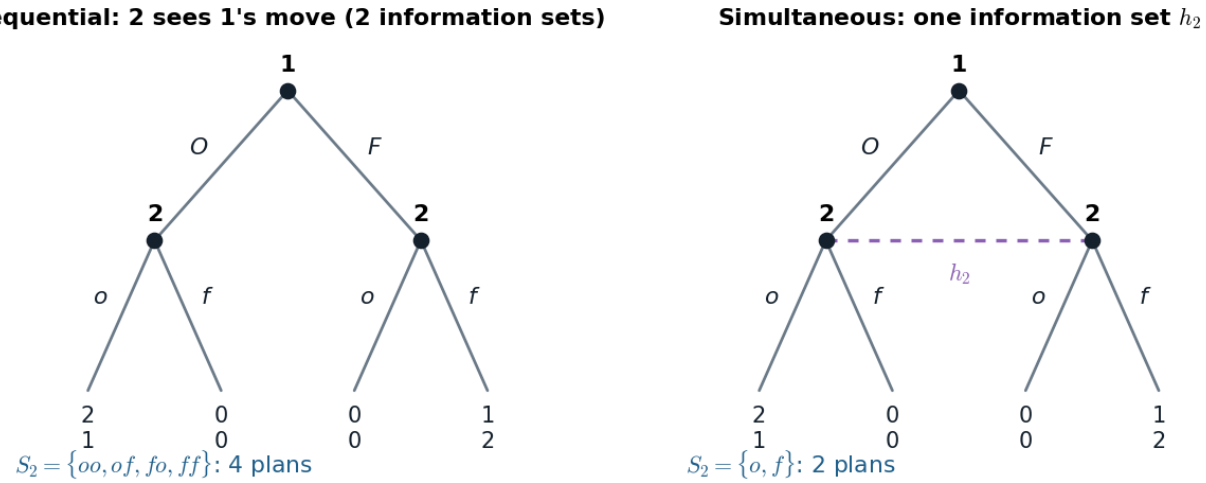


Figure — the Battle of the Sexes with the same moves and payoffs, two information structures. Left: player 2 has two information sets ($\{x_1\}, \{x_2\}$). Right: one, $h_2 = \{x_1, x_2\}$. Source: Tadelis Figs. 7.7–7.8, p. 138; payoffs p. 129.

The two trees are identical except for one dashed line, and the dashed line changes everything that follows: on the left player 2 can condition on O versus F, so she has $2 \times 2 = 4$ plans; on the right she cannot, so she has 2, and the game *is* the simultaneous Battle of the Sexes of Lecture 2.

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5 · C — perfect versus imperfect information (Definition 7.3, p. 136)

Definition 7.3. A game of complete information in which **every information set is a singleton and there are no moves of Nature** is a game of **perfect information**. A game in which some information set contains several nodes, or in which Nature moves, is a game of **imperfect information**.

Game	Information sets	Nature?	Type
Trust game, Fig. 7.1	all singletons	no	perfect
Sequential BoS, Fig. 7.2	all singletons	no	perfect
Simultaneous BoS, Fig. 7.4	$h_2 = \{x_1, x_2\}$	no	imperfect
Rock–paper–scissors, Fig. 7.5	player 2's three nodes in one set	no	imperfect

Card game, Fig. 7.6	player 1's two nodes in one set	yes	imperfect

Every simultaneous-move game is a game of imperfect information (p. 136). "Complete" (everyone knows the payoffs) and "perfect" (everyone sees every earlier move) are different properties; Lecture 3 is complete information throughout, perfect or imperfect.

5.1 · Worked — the card game (Fig. 7.6, pp. 136–137)

Nature draws a king (K) or an ace (A), each with probability $\frac{1}{2}$. Player 1 does not see it, and calls (C) or folds (F). Fold: player 1 pays 1 to player 2. Call: player 1 pays 2 on a king, receives 2 on an ace.

Player 1's action	K (prob. ½)	A (prob. ½)	Expected v_1
C	-2	$+2$	$\frac{1}{2}(-2) + \frac{1}{2}(2) = 0$
F	-1	-1	-1

Player 1 calls. The game can be drawn with Nature first (player 1's two nodes in one information set) or with player 1 first and Nature after; **both trees carry the same information, so they are the same game** (p. 137). Player 2 has no move, so this is a single-person decision problem in game clothing.

> **Two kinds of uncertainty, one device.** Not knowing Nature's draw (*exogenous* uncertainty) and not knowing another player's move (*endogenous* uncertainty) are both represented as not knowing where you are in the tree; in both cases the player needs a belief about which node she is at (p. 137). That belief is the object of Lecture 8 (Lecture 8).

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6 · G — the formal version (MWG §7.C, pp. 221–225)

MWG build the same object in their own notation: nodes, a predecessor function, a map from nodes to information sets $H(x)$, and payoffs at terminal nodes. Their Matching Pennies **Version B** (player 2 sees player 1's coin, Example 7.C.1) and **Version C** (she does not, Example 7.C.3) are the pair of trees in the figure above with different payoffs. MWG also define **perfect recall** formally (a player never forgets a move she made or saw); Tadelis states it in words as Definition 7.7 (p. 142), used in L03-02-strategies §4. Not examinable beyond the definitions above.

Links

- Next: what a strategy is once players can condition on what they saw, L03-02-strategies.
- The simultaneous Battle of the Sexes and its three equilibria: Lecture 2.
- Rules pack: rules/game-theory item 1 (*specify the game completely: players, actions, timing, information sets, payoffs*).
- Exercises (number and page only): Tadelis 7.2 (p. 147), 7.3 (p. 147–148), 7.9 (p. 149–150); Muñoz-Garcia & Toro-González Ch. 4, Exercise 3 (p. 160) and Exercise 4 (p. 163).

7.2.1–7.2.2 · A strategy is a complete plan: pure, mixed and behavioral

Claim. Once a player can condition on what she has seen, a strategy stops being an action and becomes **a complete contingent plan: one action at every information set, including the ones the plan itself makes unreachable.** Counting them is a product rule. Randomizing can happen over whole plans (mixed) or node by node (behavioral), and with perfect recall the two are equivalent (Kuhn). Prev: L03-01-extensive-form · Next: L03-03-normal-form-and-paths · Map: L03-00-map

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1 · C — pure strategies (Definition 7.4, p. 139)

Pure strategy in extensive-form games (p. 138). A complete plan of play that describes which pure action player i will choose at each of his information sets.

Definition 7.4. Let H_i be the information sets of player i, $A_i(h_i)$ the actions at h_i, and $A_i = \bigcup_{h_i \in H_i} A_i(h_i)$. A pure strategy is a mapping $s_i : H_i \to A_i$ with $s_i(h_i) \in A_i(h_i)$ for every $h_i \in H_i$. S_i is the set of all such maps.

Worked — the sequential Battle of the Sexes (Fig. 7.7, p. 138). Player 1 has one information set, so $S_1 = \{O, F\}$. Player 2 has two, $h_2^O = \{x_1\}$ (after O) and $h_2^F = \{x_2\}$ (after F), each with $\{o, f\}$. A strategy "ab" reads "play a after O, play b after F":

s_2	after O	after F	in words
oo	o	o	always opera
of	o	f	**follow player 1**
fo	f	o	do the opposite
ff	f	f	always football

$$S_2 = \{oo,\ of,\ fo,\ ff\} \qquad (p.\ 139)$$

Two actions became four strategies because there are two information sets. In the simultaneous version (one information set) $S_2 = \{o, f\}$: the same two actions, two strategies.

1.1 · Counting (p. 139)

If player i has k information sets with $m_1, \ldots, m_k$ actions:

$$\boxed{|S_i| = m_1 \times m_2 \times \cdots \times m_k}$$

because a strategy picks one action independently at each information set, and the choices multiply.

Player	Information sets and actions	$\lvert S_i \rvert$
Player 2, sequential BoS	2 sets × 2 actions	$2 \times 2 = 4$
Player 2, simultaneous BoS	1 set × 2 actions	2
Tadelis's example (p. 139)	3 sets with 2, 3, 4 actions	$2 \times 3 \times 4 = \mathbf{24}$

Player	Information sets and actions	$\lvert S_i \rvert$
Player 1, voluntary BoS (Fig. 8.4)	x_0: $\{Y, N\}$, x_1: $\{O, F\}$	$2 \times 2 = 4$: YO, YF, NO, NF
Each player, MAD game (Fig. 8.11)	2 sets $\times$ 2 actions	4

> **A strategy specifies actions at information sets the strategy itself never reaches.** In the voluntary BoS, NO and NF are different strategies even though N ends the game before O or F could be played (Tadelis p. 158; MWG p. 274, footnote 4). This is not pedantry: it is exactly what lets L03-05-subgame-perfection test whether a plan is still rational at nodes that are not reached.

Why the plan must be complete (Gibbons p. 117–118). For player j to compute a best response to i's strategy, j may need to know how i would act in *every* contingency, not only the ones expected to arise. A strategy that left some contingency blank could not be tested against deviations that lead there.

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2 · C — mixed strategies (Definition 7.5, p. 139)

> **Definition 7.5.** A mixed strategy for player i is a probability distribution over his pure strategies $s_i \in S_i$.

The player draws one complete plan **before the game** and then follows it. For player 2 in the sequential BoS a mixed strategy is $(p_{oo}, p_{of}, p_{fo}, p_{ff})$ with each $p \geq 0$ and the four summing to 1.

The interpretation has a drawback: the randomization is once-and-for-all, so the player cannot "cross the bridge when he gets there" (p. 140).

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3 · C — behavioral strategies (Definition 7.6, p. 140)

> **Definition 7.6.** A behavioral strategy specifies for each information set $h_i \in H_i$ an independent probability distribution over $A_i(h_i)$, written $\sigma_i : H_i \to \Delta A_i(h_i)$, where $\sigma_i(a_i(h_i))$ is the probability that i plays $a_i(h_i)$ at h_i.

The library analogy (Luce and Raiffa, p. 140): a pure strategy is an instruction manual with one page per information set; S_i is the library of manuals. **Mixed:** pick a manual at random, then obey it. **Behavioral:** one manual whose pages say "toss a coin here".

3.1 · Worked — mixed to behavioral (p. 141)

Given $(p_{oo}, p_{of}, p_{fo}, p_{ff})$, the probability of each action *conditional on reaching* each information set is the total weight of the plans that prescribe it there:

$$\Pr\{o \mid O\} = p_{oo} + p_{of}, \qquad \Pr\{f \mid O\} = p_{fo} + p_{ff},$$
$$\Pr\{o \mid F\} = p_{oo} + p_{fo}, \qquad \Pr\{f \mid F\} = p_{of} + p_{ff}.$$

Each pair sums to 1 because the four p's do. Setting $\sigma_2(a \mid h) = \Pr\{a \mid h\}$ gives a behavioral strategy with the same distribution over outcomes. **Every mixed strategy has a behavioral equivalent.**

3.2 · Worked — behavioral to mixed, and why it is not unique (pp. 141–142)

Fig. 7.9's behavioral strategy: $\sigma_2(o \mid O) = \frac{1}{3}$, $\sigma_2(o \mid F) = \frac{1}{2}$. A mixed strategy reproduces it iff

$$p_{oo} + p_{of} = \tfrac{1}{3}, \quad p_{fo} + p_{ff} = \tfrac{2}{3}, \quad p_{oo} + p_{fo} = \tfrac{1}{2}, \quad p_{of} + p_{ff} = \tfrac{1}{2}.$$

With $p_{oo} + p_{of} + p_{fo} + p_{ff} = 1$, the second equation follows from the first and the fourth from the third (footnote 4, p. 142), so there are **three independent equations in four unknowns**: a one-parameter family of solutions. Check the book's two:

$(p_{oo}, p_{of}, p_{fo}, p_{ff})$	sum	$p_{oo} + p_{of}$	$p_{oo} + p_{fo}$
$(\frac{1}{3}, 0, \frac{1}{6}, \frac{1}{2})$	$\frac{2+0+1+3}{6} = 1$	$\frac{1}{3}$ ✓	$\frac{1}{3} + \frac{1}{6} = \frac{1}{2}$ ✓
$(\frac{1}{6}, \frac{1}{6}, \frac{1}{3}, \frac{1}{3})$	$\frac{1+1+2+2}{6} = 1$	$\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$ ✓	$\frac{1}{6} + \frac{1}{3} = \frac{1}{2}$ ✓

A general solution: pick $p_{oo} = t \in [0, \frac{1}{3}]$, then $p_{of} = \frac{1}{3} - t$, $p_{fo} = \frac{1}{2} - t$, $p_{ff} = 1 - t - (\frac{1}{3} - t) - (\frac{1}{2} - t) = \frac{1}{6} + t$. The book's two are $t = \frac{1}{3}$ and $t = \frac{1}{6}$.

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4 · C — perfect recall and Kuhn's equivalence (Definition 7.7, p. 142)

> **Definition 7.7.** A game of **perfect recall** is one in which no player ever forgets information that he previously knew (including his own earlier moves).
>
> **Kuhn (1953), as stated on p. 142.** In games of perfect recall, mixed and behavioral strategies are equivalent: given the opponents' strategies, the same distribution over outcomes can be generated by either.

Everything in this course assumes perfect recall, so from here on "a mixed strategy" in an extensive form can be read as a behavioral one, which is how L03-05-subgame-perfection states Definition 8.3.

4.1 · U — what breaks without perfect recall: the absent-minded driver (Remark, pp. 142–143)

A driver passes two identical exits and cannot tell which one he is at. Exit 1 pays 0, exit 2 pays 4, missing both pays 1. He has **one** information set containing both intersections, so any plan is "exit with probability p whenever at an intersection":

$$V(p) = 0 \cdot p + 4\,p(1-p) + 1 \cdot (1-p)^2 = 4p - 4p^2 + 1 - 2p + p^2 = -3p^2 + 2p + 1.$$

$$V'(p) = -6p + 2 = 0 \;\Rightarrow\; p^* = \tfrac{1}{3}, \qquad V(\tfrac{1}{3}) = -\tfrac{1}{3} + \tfrac{2}{3} + 1 = \tfrac{4}{3}.$$

No pure plan does as well: $p = 0$ gives 1 and $p = 1$ gives 0. With imperfect recall **randomizing at the node can beat every pure plan**, which never happens to a single decision maker with perfect recall. At the intersection the driver's payoff from p is $q[4p(1-p) + (1-p)^2] + (1-q)[4p + (1-p)]$, where q is his belief that he is at exit 1; this coincides with the planning problem only at $q = 1$, so the plan and the in-the-moment choice disagree. **Named only; not examinable.**

Links

- Next: put these strategy sets in a matrix and find the Nash equilibria, L03-03-normal-form-and-paths.
- Mixed strategies in normal-form games: Lecture 2.
- Strategies in repeated games are history-contingent plans of the same kind: Lecture 5.
- Python: `pure_strategies`, `n_pure_strategies`, `behavioral_from_mixed`, the driver in L03-11-python §1–2.
- Exercises: Tadelis 7.1 (p. 147), 7.2(c) (p. 147), 7.4(c) (p. 148), 7.5(b) (p. 148), 7.10 (p. 150).

7.2.3–7.3 · Back to the matrix: Nash equilibria of a tree, and the threats that hold them up

> **Claim.** Every extensive form has a unique normal form, and Nash equilibrium only needs the normal form, so **finding all Nash equilibria of a tree is a matrix exercise.** The matrix also shows the problem: a Nash equilibrium only asks for best responses *on* the path of play, so it can be held up by **what a player promises to do at a node that is never reached**, even when that promise is against her own interest there. Prev: L03-02-strategies · Next: L03-04-sequential-rationality-backward-induction · Map: L03-00-map

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1 · C — from tree to matrix (§7.2.3, pp. 143–144)

Rows are player 1's pure strategies, columns player 2's, and each cell is the payoff at the terminal node that the pair of plans reaches.

Worked — sequential BoS. $S_1 = \{O, F\}$, $S_2 = \{oo, of, fo, ff\}$ (L03-02-strategies §1). If player 1 plays O, only the *first* letter of player 2's plan is used; if F, only the second:

	oo	of	fo	ff
O	$2,1$	$2,1$	$0,0$	$0,0$
F	$0,0$	$1,2$	$0,0$	$1,2$

Row O: oo and of both answer O with $o \to (2,1)$; fo and ff answer with $f \to (0,0)$. Row F: oo, fo answer F with $o \to (0,0)$; of, ff answer with $f \to (1,2)$. **Each terminal payoff appears twice** because, for a given row, two of player 2's plans differ only at the node that row does not reach (p. 144).

> **Uniqueness runs one way only.** Every extensive form has one normal form; one normal form can come from several trees (Remark, p. 144). The matrix $\begin{smallmatrix} & C & D \\ N & 0,0 & 0,0 \\ E & 1,1 & -1,2 \end{smallmatrix}$ represents both a perfect-information tree (player 2 moves only after E) and an imperfect-information tree (player 2 moves after both N and E without seeing which), Fig. 7.12. **The normal form throws away the order of moves and the information, which is exactly what Ch. 8 will need back.**

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2 · C — the Nash equilibria (§7.3, p. 145)

Mark best responses in each column for player 1 and in each row for player 2 (Lecture 2's method):

Column	Player 1's best reply	Row	Player 2's best replies
oo	O ($2 > 0$)	O	oo, of (both 1)
of	O ($2 > 1$)	F	of, ff (both 2)
fo	O or F (both 0)		
ff	F ($1 > 0$)		

A cell is a Nash equilibrium when both marks meet:

$$\{(O, oo),\ (O, of),\ (F, ff)\} \qquad (p.\ 145)$$

(F, fo) fails: player 2's best replies to F are of, ff, not fo. (O, fo) fails for the same reason in row O.

1 \ 2	*oo*	*of*	*fo*	*ff*
O	2, 1	2, 1	0, 0	0, 0
F	0, 0	1, 2	0, 0	1, 2

boxed: the three pure Nash equilibria (O, oo), (O, of), (F, ff) shaded: the unique subgame-perfect one, (O, of)

Figure — the normal form of the sequential Battle of the Sexes: three pure Nash equilibria (boxed), one of them subgame perfect (shaded). Source: Tadelis matrix p. 143, equilibria p. 145, SPE p. 157.

Two equilibria, (O, oo) and (O, of), produce the **same outcome** (both at the opera, $(2, 1)$); they differ only in what player 2 plans after F, a node neither reaches. The third, (F, ff), produces a different outcome, $(1, 2)$, and it is the one player 1 dislikes.

---

3 · C — paths of play (Definition 7.8, p. 146)

Because each node has one predecessor, every outcome corresponds to a **unique path** from the root.

> **Definition 7.8.** Let σ^* be a Nash equilibrium profile of behavioral strategies. An information set is **on the equilibrium path** if, given σ^*, it is reached with positive probability, and **off the equilibrium path** if it is never reached.

Equilibrium	Path	x_1 (after O)	x_2 (after F)	What holds player 1 on the path
(O, oo)	$O \to o$	on	**off**: plans o	nothing needed: O is best anyway
(O, of)	$O \to o$	on	**off**: plans f	nothing needed
(F, ff)	$F \to f$	**off**: plans f	on	**the plan f at x_1**: if 1 switched to O she would get 0

> **In (F, ff) player 1 stays at F only because of what player 2 says she would do at a node that is never reached** (p. 146). Player 2's plan off the path works as a threat: "if you go to the opera, I go to the football."

3.1 · Why the threat is suspect

Suppose player 1 did play O. At x_1 player 2 compares o, worth 1, with f, worth 0. **She would play o.** The plan ff commits her to the worse action at x_1, and it costs her nothing only because x_1 is

never reached. Nash equilibrium checks best replies to the *strategies*, computed from the root, so a suboptimal action at an unreached node costs nothing and passes the test (p. 146). The same logic underlies the "incredible" threat in the Stackelberg game (L03-07-stackelberg §4) and the entry game (L03-06-credibility-entry-commitment).

> **The normal form treats every choice as made once and for all at the start, so beliefs about off-path play are never put to the test** (p. 146). That is the weakness Ch. 8 repairs.

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4 · Worked — Gibbons's version (Figs. 2.4.1, 2.4.2, 2.4.5, pp. 116–128)

Player 1 chooses L or R; player 2 sees it and chooses L' or R'.

	after L	after R
L'	$(3,1)$	$(2,1)$
R'	$(1,2)$	$(0,0)$

Player 2's four strategies are written (action after L, action after R). Normal form (Fig. 2.4.2):

	(L',L')	(L',R')	(R',L')	(R',R')
L	$3,1$	$3,1$	$1,2$	$1,2$
R	$2,1$	$0,0$	$2,1$	$0,0$

Best replies: player 2 against L wants R' first ($2>1$), i.e. (R',L') or (R',R'); against R wants L' second ($1>0$), i.e. (L',L') or (R',L'). Player 1 against (R',L'): L gives 1, R gives $2 \to R$; against (R',R'): L gives 1, R gives $0 \to L$.

$$\text{NE} = \{(R,(R',L')),\ (L,(R',R'))\} \qquad (p.\ 126)$$

In $(L,(R',R'))$ player 2 "threatens" R' after R, which would give her 0 instead of 1 from L'. **The threat is not credible, and $(L,(R',R'))$ is not subgame perfect** (p. 128). The tree is solved in L03-04-sequential-rationality-backward-induction §3.

Links

- Next: rule the empty threats out, L03-04-sequential-rationality-backward-induction.
- Best-response marking and the simultaneous BoS: Lecture 2.
- Python: `pure_nash` on `bos_seq` and `gib`, L03-11-python §1.
- Exercises: Tadelis 7.2(d) (p. 147), 7.4(d) (p. 148), 7.5(c) (p. 148), 7.6 (p. 148–149), 7.7 (p. 149), 7.8 (p. 149); Muñoz-Garcia & Toro-González Ch. 4, Exercise 4 (p. 163), Exercise 5 (p. 166), Exercise 6 (p. 169).

8.1 · Sequential rationality and backward induction

Claim. Ask every player to play a best response **at every information set, reached or not**: that is sequential rationality. In a finite game of perfect information it is enforced by solving the tree from the end, and the procedure always delivers an answer, unique when no player is ever indifferent (Zermelo, Proposition 8.1). The answer can be bleak: in the Centipede game it throws away a payoff both players would prefer. Prev: L03-03-normal-form-and-paths · Next: L03-05-subgame-perfection · Map: L03-00-map

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1 · U — a coordination game with a silly equilibrium (Fig. 8.1, p. 151)

Player 1 exits (E) with $(0, 0)$ or proceeds (P); after P, player 2 exits (e) with $(0, 0)$ or proceeds (p) with $(1, 1)$.

	e	p
E	$0, 0$	$0, 0$
P	$0, 0$	$1, 1$

Two Nash equilibria: (P, p) and (E, e). In (E, e) player 1 exits because player 2 "would exit"; but if player 2 were ever asked, p gives 1 against 0 from e. Nash equilibrium **puts no restriction on beliefs off the path**, so it lets (E, e) through (pp. 151–152).

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2 · C — the definition and the procedure (§8.1, pp. 152–153)

Definition 8.1. Given strategies $\sigma_{-i} \in \Delta S_{-i}$ of i's opponents, player i is **sequentially rational** if and only if i is playing a best response to σ_{-i} **in each of his information sets**.

Backward induction (p. 153): call "level 1" the nodes that immediately precede terminal nodes; at each, let the mover pick the action with her highest payoff. Call "level 2" the nodes that precede level-1 nodes; let the mover pick her best action *given* the level-1 choices. Continue up to the root. The game is finite, so the procedure ends; at every node the mover best-responds to what comes after, so the result is sequentially rational everywhere.

Proposition 8.1 (Zermelo's Theorem). Any finite game of perfect information has a backward induction solution that is sequentially rational. Furthermore if no two terminal nodes prescribe the same payoffs to any player then the backward induction solution is unique.

Corollary 8.1. Any finite game of perfect information has at least one sequentially rational Nash equilibrium **in pure strategies**. If no two terminal nodes prescribe the same payoffs to any player, the game has a unique sequentially rational Nash equilibrium.

Hypothesis	What fails without it
finite	there may be no last nodes to start from (Rubinstein bargaining, repeated games: Lecture 4, Lecture 5)

Hypothesis	What fails without it
perfect information	at a non-singleton information set the mover's best action depends on a belief about where she is, which the procedure does not supply (p. 154; see L03-05-subgame-perfection)
no ties (for uniqueness)	an indifferent mover can pick either action and each choice can change earlier players' choices (footnote 1, p. 153)

History (Remark, p. 153): Zermelo (1913) showed that in games like chess a player can guarantee at least a draw; neither backward induction nor strategies appear in his paper. MWG state the same result as **Proposition 9.B.1** (§9.B): every finite game of perfect information has a pure-strategy Nash equilibrium that can be derived by backward induction, unique if no player has the same payoffs at two terminal nodes.

2.1 · Worked — the sequential Battle of the Sexes (p. 152)

Step	Node	Comparison	Choice
1	x_1 (after O), player 2	o: 1 vs f: 0	o
1	x_2 (after F), player 2	o: 0 vs f: 2	f
2	x_0, player 1	$O \to (2,1)$ vs $F \to (1,2)$: 2 vs 1	O

Result: player 1 plays O, player 2 plays of. (O, of) **is the only one of the three Nash equilibria of L03-03-normal-form-and-paths that survives** (p. 152). (O, oo) dies because o at x_2 is not a best response there; (F, ff) dies because f at x_1 is not.

> **Backward induction predicts play off the path as well as on it**: "if player 1 chooses F then player 2 will choose f" (p. 152). That off-path prediction is part of the answer.

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3 · Worked — Gibbons's games (§2.1.A, pp. 57–61; §2.4.B, p. 126)

Two stages (Gibbons pp. 57–58). Player 1 picks a_1; player 2 sees it and solves $\max_{a_2} u_2(a_1, a_2)$, with solution $R_2(a_1)$; player 1 anticipates this and solves $\max_{a_1} u_1(a_1, R_2(a_1))$, with solution a_1^*. $(a_1^*, R_2(a_1^*))$ **is the backwards-induction outcome.** It "does not involve noncredible threats: player 1 gives no credence to threats by player 2 to respond in ways that will not be in 2's self-interest when the second stage arrives" (p. 58).

Fig. 2.4.1 solved (p. 126; matrix in L03-03-normal-form-and-paths §4): after L, player 2 prefers R' ($2 > 1$); after R, L' ($1 > 0$). Player 1 compares $L \to 1$ with $R \to 2$ and plays R. Outcome (R, L'); equilibrium $(R, (R', L'))$.

Three moves, player 1 moves twice (pp. 59–60). L ends with $(2, 0)$. After R, player 2 plays L' $(1, 1)$ or R'; after R', player 1 plays L'' $(3, 0)$ or R'' $(0, 2)$.

Step	Mover	Comparison	Choice
3	player 1 at stage 3	L'': 3 vs R'': 0	L''

Step	Mover	Comparison	Choice
2	player 2	L': 1 vs R' then L'': 0	L'
1	player 1	L: 2 vs R then L': 1	L

The game ends at once, at $(2, 0)$. **Gibbons's point** (pp. 60–61): the argument depends on what player 2 would believe *after* player 1 has played R, a move backward induction says will not happen. If 1 played R, it cannot be common knowledge that both are rational, yet there are reasons for R consistent with 1 being rational (for instance, 1 doubts that 2 is rational and hopes for R'). Backward induction assumes deviations are explained that way; where it is more plausible that a deviator is simply irrational, it loses appeal.

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4 · Worked — the Centipede Game (§8.3.1, Fig. 8.8, pp. 159–160)

Players alternate; each can stop (N for 1, n for 2) or continue (C, c). Payoffs if stopped at each node, and at the end:

Node	Mover	Stop gives	Continue leads to
1	1	$(1, 1)$	node 2
2	2	$(0, 3)$	node 3
3	1	$(2, 2)$	node 4
4	2	$(1, 4)$	end $(3, 3)$

Backward induction:

Step	Node	Mover compares	Choice	Value carried back
1	4	player 2: $n \to 4$ vs $c \to 3$	n	$(1, 4)$
2	3	player 1: $N \to 2$ vs $C \to 1$	N	$(2, 2)$
3	2	player 2: $n \to 3$ vs $c \to 2$	n	$(0, 3)$
4	1	player 1: $N \to 1$ vs $C \to 0$	N	$(1, 1)$

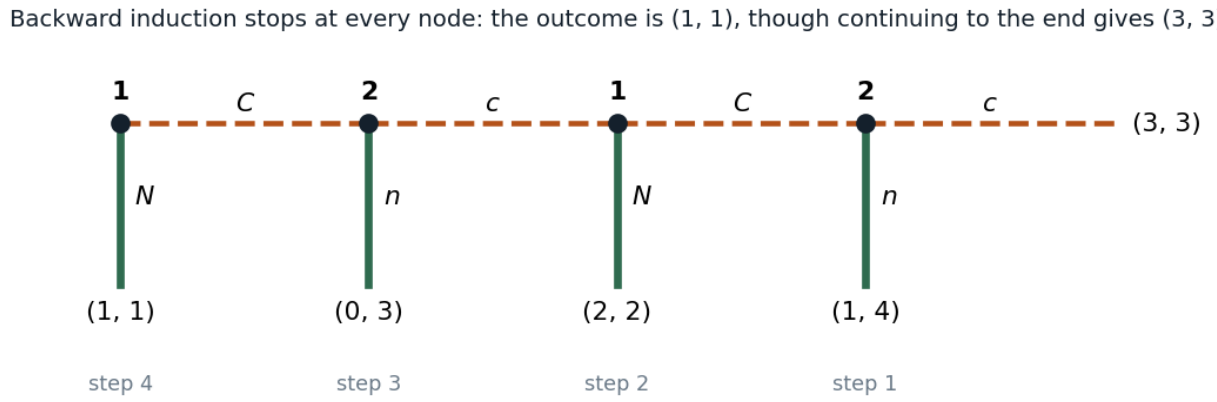


Figure — backward induction in the Centipede Game: every mover stops, so play ends at once with $(1, 1)$ *although* $(3, 3)$ *is on the table. Source: Tadelis Fig. 8.8, p. 159.*

Read the figure right to left: each step compares one stop payoff with the value that the next step has already carried back, and stopping wins every time by exactly 1.

> **The sum of payoffs rises by 1 at every continuation, so the structure can be stretched to make the end as attractive as one likes; the unique backward induction outcome stays** $(1, 1)$ (p. 159–160).

Evidence (Remark, p. 160): in the lab, people mostly do not stop at once (McKelvey and Palfrey, 1992). Two explanations: players care about each other's payoffs, or rationality is not common knowledge. Palacios-Huerta and Volij (2009) ran it with chess players: 69% stopped immediately, and grandmasters always did, which supports the second explanation.

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5 · G — MWG's framing (§9.B, Examples 9.B.2 and 9.B.5)

MWG motivate the same procedure with the *principle of sequential rationality*: "a player's strategy should specify optimal actions at every point in the game tree" (p. 269), and solve a three-player perfect-information game by successive reduction (Example 9.B.2). Their Example 9.B.5 is a 100-stage centipede. Not examinable beyond the statement of Proposition 9.B.1.

Links

- Next: what to do when the end of the tree contains a non-singleton information set, L03-05-subgame-perfection.
- The same logic in finite bargaining and finitely repeated games: Lecture 4, Lecture 5.
- Rules pack: rules/game-theory item 7 (*backward induction for finite games of perfect information*).
- Python: `backward_induction` on `bos_seq`, `gib3`, `cent`, L03-11-python §1.
- Practice: Listas/pratica-aula-01-02.md item 19 (Cabral 7.14, Centipede, p. 182).
- Exercises: Tadelis 7.4 (p. 148), 7.5 (p. 148), 8.11(b) (p. 172–173); Muñoz-Garcia & Toro-González Ch. 4, Exercise 9 (p. 176), Exercise 19 (p. 202), Exercise 20 (p. 204).

8.2–8.3.3 · Subgame-perfect equilibrium: backward induction for any tree

> **Claim.** Backward induction stalls at a non-singleton information set, because the mover's best action depends on a belief the procedure does not supply. The fix is to cut the tree only where it can be cut cleanly, at **proper subgames**, and to require a **Nash equilibrium in every one of them**. That is subgame perfection (Selten). It refines Nash, coincides with backward induction under perfect information, and **can still leave several equilibria when a subgame has several equilibria of its own**, as mutually assured destruction shows. Prev: L03-04-sequential-rationality-backward-induction · Next: L03-06-credibility-entry-commitment · Map: L03-00-map

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1 · U — where backward induction gets stuck (Fig. 8.2, p. 154)

The voluntary Battle of the Sexes. Player 1 chooses N (both get 1.5) or Y; after Y the two play the *simultaneous* Battle of the Sexes. Player 2's two nodes sit in one information set. To start backward induction we would need player 2's best action at that set, and it depends on whether she thinks player 1 chose O or F: a belief, which backward induction does not provide (p. 154).

The second obstacle is games with no last node (infinite horizons, Chs. 10–11), which have no terminal nodes to start from. Subgame perfection handles both.

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2 · C — proper subgames (Definition 8.2, pp. 154–155)

> **Definition 8.2.** A proper subgame G of an extensive-form game Γ consists of **a single node and all its successors** in Γ, with the property that **if** $x \in G$ **and** $x' \in h(x)$ **then** $x' \in G$. The subgame is itself a game tree, with information sets and payoffs inherited from Γ.

Two tests for "can a subgame start at node x?":

1. x **is a singleton information set** (otherwise the subgame would have no unique root);
2. **no information set straddles the cut**: every information set that contains a successor of x lies entirely among the successors of x.

Game	Candidate roots	Proper subgames	Why
Perfect information (Fig. 8.3, p. 155)	every decision node	**one per decision node**	all information sets are singletons
Simultaneous BoS	root only	**the whole game**	player 2's nodes share an information set
Voluntary BoS (Fig. 8.4)	x_0, x_1; not x_2, x_3	**two**: the whole game and the one at x_1	x_2, x_3 are in one set, so neither can be a root (p. 156)
Card game (Fig. 8.5, p. 156)	x_1, x_2 fail test 2	**the whole game only**	the subgame at x_1 would contain x_3 and so, by the definition, $x_4 \in h(x_3)$, which x_1 does not precede

Game	Candidate roots	Proper subgames	Why
MAD (Fig. 8.11)	root, player 2's node, player 1's war-stage node	**three**	the war stage is simultaneous but starts at a singleton (p. 165)

Gibbons's Definition (pp. 122–123) adds a parenthetical that the whole game is *not* counted as a subgame; he notes this is "only a matter of style". Tadelis and MWG (Definition 9.B.1) count the whole game as a subgame. These notes follow Tadelis.

> **Why the second test.** Within a subgame, a player's best response should depend only on what others do inside it (p. 156). Gibbons's reason (pp. 123–124): test 2 guarantees that the history up to x is **common knowledge** among everyone who moves later in the subgame, so the subgame can be analysed on its own and the analysis is relevant to the original game.

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3 · C — the equilibrium (Definition 8.3, p. 157)

> **Definition 8.3 (Selten 1975).** Let Γ be an n-player extensive-form game. A behavioral strategy profile $\sigma^* = (\sigma_1^*, \dots, \sigma_n^*)$ is a **subgame-perfect (Nash) equilibrium** if for every proper subgame G of Γ the restriction of σ^* to G is a Nash equilibrium in G.

Consequences, each one line:

Statement	Why
Every SPE is a Nash equilibrium	the whole game is a proper subgame
Not every Nash equilibrium is an SPE	e.g. (F, ff), (O, oo) in the sequential BoS (p. 157)
If the only proper subgame is the whole game, SPE = NE	nothing else to check (MWG Exercise 9.B.2(a)); the card game and every simultaneous game
Fact (p. 158): in finite games of perfect information, **SPE = the profiles that survive backward induction**	every node roots a subgame, and a Nash equilibrium of a one-mover subgame is that mover's best action
At least one SPE exists in finite games (Summary, p. 169)	Gibbons p. 124: each finite subgame has a Nash equilibrium (Nash's theorem), and the reduction below builds one

> **Subgame perfection requires best responses off the equilibrium path as well as on it**: the restriction must be a Nash equilibrium "in every proper subgame, including those that are not reached in equilibrium" (p. 157).

Sequential BoS (p. 157–158): the subgames at x_1 and x_2 have one player each, so their only Nash equilibria are o at x_1 and f at x_2. Of $\{(O, oo), (O, of), (F, ff)\}$ only (O, of) has those restrictions.

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4 · C — the method: solve the last subgames, replace, repeat

Gibbons's footnote 20 (p. 124) and MWG Proposition 9.B.3 give the procedure:

1. Find the **smallest** subgames (those containing no other subgame).
2. Find a Nash equilibrium of each, pure or mixed.

3. **Replace** each such subgame by a terminal node carrying that equilibrium's payoffs.
4. Repeat on the reduced game until the root.
5. If a subgame has several equilibria, **branch**: run the rest once for each.

MWG Proposition 9.B.3 (G) is the justification: (i) in any SPE that plays σ_S in subgame S, the moves outside S form an SPE of the reduced game; (ii) any SPE of the reduced game, completed by σ_S inside S, is an SPE of the whole game.

4.1 · Worked — the voluntary Battle of the Sexes (pp. 158–159)

Strategies: $S_1 = \{YO, YF, NO, NF\}$ (NO and NF differ only at x_1, which N never reaches), $S_2 = \{o, f\}$. Normal form:

	o	f
YO	$2, 1$	$0, 0$
YF	$0, 0$	$1, 2$
NO	$1.5, 1.5$	$1.5, 1.5$
NF	$1.5, 1.5$	$1.5, 1.5$

Nash equilibria. Column o: player 1's best reply is YO ($2 > 1.5 > 0$); row YO: player 2 plays o ($1 > 0$) $\to (YO, o)$. Column f: player 1's best replies are NO, NF ($1.5 > 1 > 0$); rows NO, NF: player 2 is indifferent, so f is a best reply $\to (NO, f)$, (NF, f). Row YF: player 2 plays f, but then player 1 prefers $N \to$ not an equilibrium.

$$E_{\text{Nash}} = \{(YO, o),\ (NO, f),\ (NF, f)\} \qquad (p.\ 158)$$

Subgame perfection. The subgame at x_1 is the simultaneous BoS, whose pure equilibria are (O, o) and (F, f) (and one mixed, left to Tadelis exercise 8.1, p. 170). Branch:

Equilibrium in the subgame	Payoffs carried up	Player 1 at x_0	SPE
(O, o)	$(2, 1)$	Y: 2 vs N: 1.5 $\to Y$	(YO, o)
(F, f)	$(1, 2)$	Y: 1 vs N: 1.5 $\to N$	(NF, f)

$$E_{\text{SPE}} = \{(YO, o),\ (NF, f)\} \qquad (p.\ 158)$$

(NO, f) **is a Nash equilibrium but not subgame perfect**: restricted to the subgame at x_1 it is (O, f), which is not an equilibrium of the simultaneous BoS (player 2 would switch to o). It survives the Nash test only because x_1 is never reached.

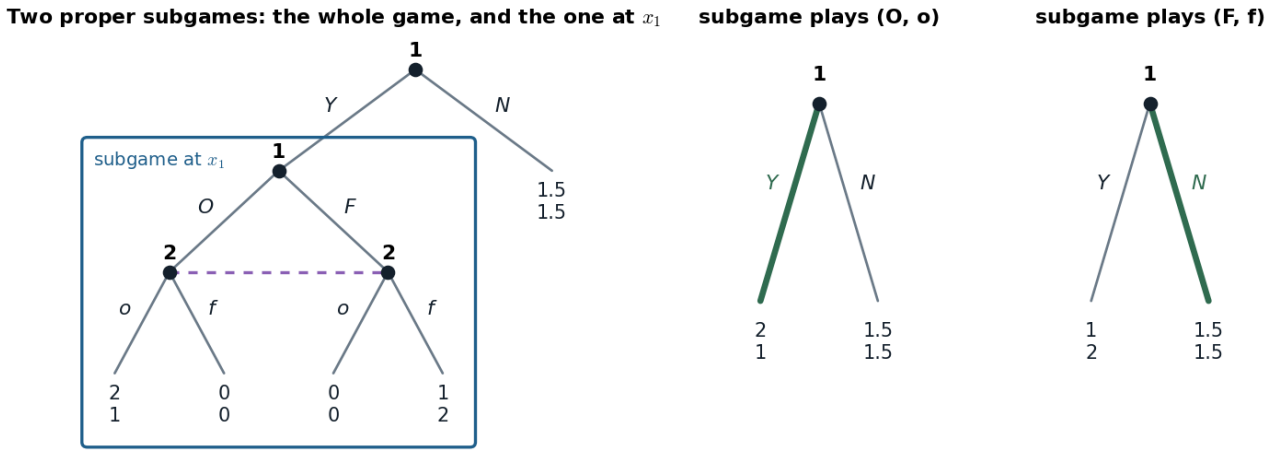


Figure — the two proper subgames of the voluntary BoS and the two reduced games, one per pure equilibrium of the subgame. Source: Tadelis Figs. 8.4 and 8.7, pp. 155, 159.

The box is the only place a subgame can start below the root; the dashed line inside it is why neither of player 2's nodes can start one. Each right-hand panel is the whole game after the box has been replaced by one of its equilibria, and player 1's choice flips between them because 1.5 sits between the two payoffs 2 and 1.

4.2 · Worked — mutually assured destruction (§8.3.3, Fig. 8.11, pp. 163–166)

Player 1 (the United States) ignores (I, payoffs $(0, 0)$) or escalates (E). After E, player 2 (the USSR) backs down (B, $(10, -10)$) or goes nuclear (N). After N they play a simultaneous **war stage**: retreat (R, r) or Doomsday (D, d).

	r	d
R	$-5, -5$	$-100, -100$
D	$-100, -100$	$-100, -100$

Strategies: $S_1 = \{IR, ID, ER, ED\}$, $S_2 = \{Br, Bd, Nr, Nd\}$. The pure Nash equilibria of the whole game (p. 164):

$$E_{\text{Nash}} = \{(IR, Nr), (IR, Nd), (ID, Nr), (ID, Nd), (ED, Br), (ED, Bd)\}$$

Step 1, the war stage. (R, r): neither gains by switching ($-5 > -100$). (D, d): either player switching alone still gets -100, so no strict gain: also an equilibrium. Two pure Nash equilibria: NE1 $= (R, r)$ with $(-5, -5)$, NE2 $= (D, d)$ with $(-100, -100)$ (p. 165).

Step 2, branch on the war stage (Fig. 8.12, pp. 165–166):

Case	War stage worth	Player 2: N vs B	Player 1: E vs I	SPE
1: NE1	$(-5, -5)$	$-5 > -10 \to N$	$E \to -5$ vs $I \to 0 \to I$	(IR, Nr)
2: NE2	$(-100, -100)$	$-10 > -100 \to B$	$E \to 10$ vs $I \to 0 \to E$	(ED, Bd)

$$E_{\text{SPE}} = \{(IR, Nr),\ (ED, Bd)\} \qquad (p.\ 166)$$

Why the other four Nash equilibria fail, each in the first subgame where its restriction is not a Nash equilibrium:

Nash equilibrium	War stage played	Fails because
(IR, Nd)	(R, d)	player 2 switches to r: $-5 > -100$
(ID, Nr)	(D, r)	player 1 switches to R: $-5 > -100$
(ED, Br)	(D, r)	player 1 switches to R: $-5 > -100$
(ID, Nd)	(D, d), an equilibrium	at player 2's node, B gives $-10 > -100$ from N followed by (D, d)

> **In both SPE the war stage is off the path, and still it decides everything** (p. 166): what the players expect to happen in a subgame nobody reaches fixes who escalates and who backs down. The escalation equilibrium (ED, Bd) works because both expect Doomsday if war comes.

The mixed equilibrium of the war stage and the SPE built on it are Tadelis exercise 8.2 (p. 170).

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5 · C — two-stage games with simultaneous moves (Gibbons §2.2.A, pp. 71–73)

Gibbons's class: players 1 and 2 move simultaneously; players 3 and 4 observe (a_1, a_2) and then move simultaneously. Assume each second-stage game has a unique Nash equilibrium $(a_3^*(a_1, a_2), a_4^*(a_1, a_2))$. Then the first stage is the simultaneous game with payoffs $u_i(a_1, a_2, a_3^*(a_1, a_2), a_4^*(a_1, a_2))$; if its Nash equilibrium (a_1^*, a_2^*) is unique, the **subgame-perfect outcome** is $(a_1^*, a_2^*, a_3^*(a_1^*, a_2^*), a_4^*(a_1^*, a_2^*))$.

Outcome versus equilibrium (Gibbons pp. 125–126): the *outcome* names the moves on the path; the *equilibrium* names complete strategies, $(a_1^*, a_2^*, a_3^*(\cdot,\cdot), a_4^*(\cdot,\cdot))$, a reply for every first-stage pair. In the two-stage perfect-information game the outcome is $(a_1^*, R_2(a_1^*))$ and the SPE is $(a_1^*, R_2(\cdot))$. **Writing the outcome when an equilibrium is asked for is the most common error on this topic** (L03-10-pitfalls §1).

This class is the template for the multistage games of Lecture 4 (bank runs, tariffs, tournaments are Gibbons's §2.2.B–D).

Links

- Next: the economic use of the refinement, credibility and commitment, L03-06-credibility-entry-commitment.
- Mixed equilibria of a subgame: Lecture 2. Infinite-horizon SPE: Lecture 4, Lecture 5. Beliefs at information sets (perfect Bayesian equilibrium): Lecture 8.
- Rules pack: rules/game-theory item 2 (*Nash equilibrium in an extensive game admits incredible threats that subgame perfection removes*), item 3 (multiplicity).

- Python: `subgame_roots`, `spe` on `vbos`, `card`, `mad`, L03-11-python §1.
- Exercises: Tadelis 8.1 (p. 170), 8.2 (p. 170), 8.3 (p. 170), 8.5 (p. 170), 8.10 (p. 172); Muñoz-Garcia & Toro-González Ch. 4, Exercise 4 (p. 163), Exercise 6 (p. 169).

Applications · Credible threats, entry deterrence and the value of commitment

Claim. An incumbent's threat to fight an entrant can hold up a Nash equilibrium and still be empty, because once entry has happened fighting pays the incumbent less than sharing. Subgame perfection deletes that equilibrium. **The incumbent's remedy is to change its own payoffs before the entrant moves**, so that fighting becomes its best reply: a credible commitment. Cabral's numbers put the value of that commitment at 30, and it is earned by making one of the incumbent's own payoffs *worse*. Prev: L03-05-subgame-perfection · Next: L03-07-stackelberg · Map: L03-00-map

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1 · C — the sequential-entry game (Cabral Fig. 7.7, p. 168)

A monopolized industry. Player 1 (the **entrant**) chooses e (enter) or $\bar{e}$ (stay out). If it enters, player 2 (the **incumbent**) chooses r (retaliate: a price war) or $\bar{r}$ (accommodate). Payoffs (entrant, incumbent):

Path	Entrant	Incumbent
$\bar{e}$ (stay out)	0	50
e, $\bar{r}$ (accommodate)	10	20
e, r (retaliate)	-10	10

Source detail. Fig. 7.7 prints $(-10, 10)$ after a price war; the text on p. 168 says the incumbent "would get -10" there. Every conclusion below holds for either value, since both are below 20 (the incumbent prefers to accommodate) and above -20 (the bonded case of §3). The companion lets you check both.

1.1 · Nash equilibria (p. 168)

	$\bar{r}$	r
e	$10, 20$	$-10, 10$
$\bar{e}$	$0, 50$	$0, 50$

- $(e, \bar{r})$: given e, the incumbent gets 20 from $\bar{r}$ against 10 from r; given $\bar{r}$, the entrant gets 10 from e against 0. **Nash.**
- $(\bar{e}, r)$: given r, the entrant gets 0 from $\bar{e}$ against -10 from e; given $\bar{e}$, the incumbent gets 50 whatever it plans, so r is a best reply ("though not the only one"). **Nash.**

1.2 · Only one survives (p. 169)

The second equilibrium rests on the incumbent's plan to retaliate at a node that $\bar{e}$ never reaches. **Is the threat credible?** If the entrant did enter, the incumbent would compare 10 (retaliate) with 20 (accommodate) and accommodate. Backward induction:

Step	Node	Comparison	Choice
1	incumbent after e	$\bar{r}$: 20 vs r: 10	$\bar{r}$
2	entrant	e then $\bar{r}$: 10 vs $\bar{e}$: 0	e

$$\text{SPE} = (e, \bar{r}), \quad \text{payoffs } (10, 20).$$

For a tree this simple, subgame perfection and backward induction coincide (p. 169), as the Fact of L03-05-subgame-perfection §3 says.

1.3 · The same game in MWG (Example 9.B.1, pp. 268–269)

MWG's predation game has out $(0, 2)$, in–fight $(-3, -1)$, in–accommodate $(2, 1)$, and the same two Nash equilibria, (out, fight if in) and (in, accommodate if in). Their diagnosis is the clearest statement of the problem:

> "When the entrant plays 'out,' actions at decision nodes that are unreached by play of the equilibrium strategies … do not affect firm I's payoff. As a result, firm I can plan to do absolutely anything at this decision node … But … what firm I's strategy says it will do at the unreached node can actually insure that firm E, taking firm I's strategy as given, wants to play 'out.'" (p. 269)

Their Example 9.B.3 (§9.B) replaces the post-entry move by a *simultaneous* fight/accommodate game between both firms; it has three pure Nash equilibria, and solving the post-entry subgame first (accommodate, accommodate) leaves a unique SPE in which E enters. (G layer.)

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2 · U — why "threat" is the right word, and why it fails

Gibbons (pp. 127–128), on the equilibrium $(L, (R', R'))$ of his Fig. 2.4.1 (L03-03-normal-form-and-paths §4): "one might say that player 2 is threatening to play R' if player 1 plays R … If this threat works … then 2 is not given the opportunity to carry out the threat. The threat should not work, however, because it is not credible."

	What it is	What the entrant does with it
Threat	a plan to act against your own interest if the other side does X	believes it in a Nash equilibrium
Credible threat	a plan that is your best reply *when the moment comes*	believes it in a subgame-perfect equilibrium
Non-credible threat	a plan you would abandon when the moment comes	ignores it

> **A threat is credible only if carrying it out is optimal at the node where it would be carried out.** Saying it louder, or saying it first, changes nothing; only changing the payoffs at that node does.

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3 · C — the value of credible commitment (Cabral pp. 169–170, Fig. 7.8)

Suppose the incumbent can first sign an **enforceable, non-renegotiable** contract (a bond): if the entrant enters and the incumbent does *not* retaliate, the incumbent pays a penalty of 40. Its accommodation payoff drops from 20 to $20 - 40 = -20$. Everything else is unchanged. The game now starts with the incumbent choosing bond (b) or no bond ($\bar{b}$).

Subgame	Incumbent after entry	Entrant	Incumbent's payoff
no bond (Fig. 7.7)	$\bar{r}$: 20 vs r: 10 $\to \bar{r}$	e: 10 vs $\bar{e}$: 0 $\to$ **enter**	20
bond	$\bar{r}$: -20 vs r: 10 $\to r$	e: -10 vs $\bar{e}$: 0 $\to$ **stay out**	50

At the root the incumbent compares 50 with 20 and **posts the bond**.

$$\text{value of commitment} = 50 - 20 = \mathbf{30} \qquad (p.\ 170)$$

Fig. 7.7: 'retaliate' is a non-credible threat

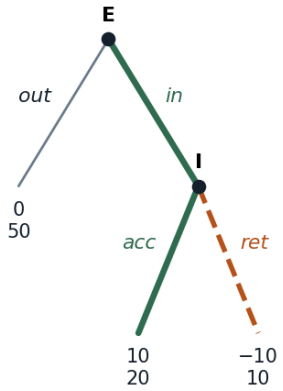


Fig. 7.8: a bond of 40 makes retaliation credible

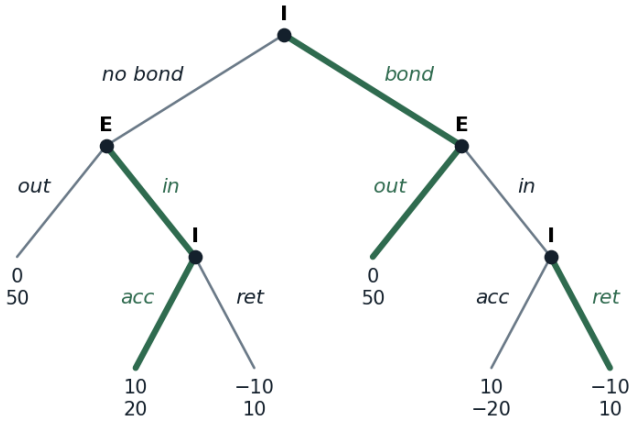


Figure — the entry game without and with a bond. Green: subgame-perfect choices at every node; dashed orange: the non-credible threat. Source: Cabral Figs. 7.7–7.8, pp. 168–170; bond 40, p. 169.

In the right-hand tree the two subgames differ in a single number, 20 versus -20 in the incumbent's accommodation payoff, and that one number flips the incumbent's post-entry choice, which flips the entrant's choice, which moves the incumbent from 20 to 50.

> **"A credible commitment may have significant strategic value"** (Cabral p. 170). The bond makes the incumbent worse off in one terminal node and better off in equilibrium, because the node it worsens is the one the threat needed it to avoid.

3.1 · How big must the bond be?

Retaliation becomes the incumbent's strict best reply after a bond B when

$$\underbrace{10}_{r} > \underbrace{20 - B}_{\bar{r}} \quad \Longleftrightarrow \quad B > B^{*} = 20 - 10 = 10.$$

B	Accommodation pays	Incumbent after entry	Entrant	Incumbent's payoff
0	20	accommodate	enter	20
5	15	accommodate	enter	15 (worse than no bond: not posted)

B	Accommodation pays	Incumbent after entry	Entrant	Incumbent's payoff
10	10	**indifferent**: two SPE	enter or stay out	10 or 50
11	9	retaliate	stay out	50
40 (Cabral)	-20	retaliate	stay out	50

A bond below B^* is worse than none: it lowers the accommodation payoff without changing behaviour. With the text's value of -10 for retaliation, $B^* = 20 - (-10) = 30$ and Cabral's 40 still clears it.

> **Interactive companion.** The Credible Threat Desk (`companions/companion-entry-commitment.html`) — move the bond through B^* and watch the incumbent's subgame-perfect payoff jump from 20 to 50; move the incumbent's war payoff above its accommodation payoff and the bond becomes worthless because the threat is credible on its own; push the entrant's war payoff above 0 and no threat deters it.

3.2 · Two ways to model commitment (pp. 170–171)

Cabral's methodological point: if a player is credibly committed to an action, model it either by **changing its payoffs** (the bond, Fig. 7.8) or by **changing the order of moves**, letting it choose r or $\bar{r}$ *before* the entrant moves (Fig. 7.9). Solving Fig. 7.9 backward gives the same answer as Fig. 7.8. What a commitment does, in game-theoretic terms, is **turn a later move into an earlier one**. The Stackelberg leader of L03-07-stackelberg is exactly this: a firm whose quantity is fixed before its rival moves.

3.3 · Short-run and long-run variables (p. 171, Fig. 7.10)

When firms choose both a long-run variable (capacity, plant, product position) and a short-run one (price), model the long-run choice in the **first stage** and the short-run choice in the second: short-run variables are chosen *given* the long-run ones. The two-stage game is then solved by subgame perfection, the second stage first. Cabral uses this timing for capacity and price, entry and output, and product positioning and price in his Chs. 8, 10 and 14.

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4 · Organizational reading

Setting	The non-credible threat	The commitment that fixes it
Incumbent vs entrant	"we will start a price war"	sunk capacity, a bond, a reputation at stake in other markets
Seller vs shopper	"the price will be fair once you are here" (after the trip is sunk, the seller takes everything)	a price posted in advance (Tadelis exercise 8.13, p. 173–174)
Lender vs debtor	"we will sue if you default" when suing costs more than it recovers	a cheap, enforceable legal system (Tadelis exercise 8.7, p. 171)

The broader theme, **commitment as a source of value and renegotiation as its enemy**, returns

with relational contracts in Lecture 6; Cabral's footnote n (p. 169) already flags that most real contracts are renegotiable, which is what makes the bond assumption strong.

Links

- Previous: the refinement itself, L03-05-subgame-perfection. Next: commitment through a quantity, L03-07-stackelberg.
- Rules pack: rules/game-theory item 2; commitment and hold-up in contracts: rules/06.
- Python: `cabral` and `mwg` games, L03-11-python §1.
- Practice: Listas/pratica-aula-01-02.md (Cabral §7.2, pp. 167–174, marked "Exam").
- Exercises: Tadelis 7.6 (p. 148–149), 8.7 (p. 171), 8.8 (p. 171–172), 8.9 (p. 172), 8.13 (p. 173–174); Muñoz-Garcia & Toro-González Ch. 4, Exercise 2 (p. 156).

8.3.2 · Stackelberg competition: the leader's quantity is a commitment

Claim. Make Cournot sequential: firm 1 chooses q_1, firm 2 sees it and replies. Backward induction turns firm 2's best-response function into a **menu** from which firm 1 picks its favourite point: $q_1 = 45$, $q_2 = 22.5$, profits 1012.5 and 506.25 against Cournot's 900 each. **The leader could have produced its Cournot quantity and chose not to, so it gains; the follower loses; total output rises and price falls.** And the outcome is not an equilibrium: the SPE must specify firm 2's reply to *every* q_1. Prev: L03-06-credibility-entry-commitment · Next: L03-08-time-inconsistency · Map: L03-00-map

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1 · C — the game (pp. 160–161)

Tadelis's Cournot example: inverse demand $p = 100 - q$ with $q = q_1 + q_2$, costs $c_i(q_i) = 10q_i$. In the simultaneous game (Lecture 2) each firm solves $\max_{q_i}(100 - q_i - q_j)q_i - 10q_i$, with first-order condition $100 - 2q_i - q_j - 10 = 0$, so

$$q_i(q_j) = \frac{90 - q_j}{2}.$$

Change one detail: firm 1 chooses q_1 first, firm 2 **observes it** and then chooses q_2.

	Cournot (Fig. 8.10)	Stackelberg (Fig. 8.9)
Firm 2's information	one information set containing every q_1	**a separate information set for each** q_1 (a continuum)
Proper subgames	the whole game only	the whole game **and one per** q_1
Type	imperfect information	perfect information

---

2 · C — backward induction, every step (pp. 160–161)

Step 1: the follower. At the information set where it has seen q_1, firm 2 maximizes $(100 - q_1 - q_2)q_2 - 10q_2$. First-order condition: $100 - q_1 - 2q_2 - 10 = 0$, so

$$q_2(q_1) = \frac{90 - q_1}{2} \qquad (8.1)$$

The formula is Cournot's best response, but its meaning changed: there it answered a *belief* about q_1; here it answers an *observed* q_1 (p. 161; Gibbons p. 62).

Step 2: the leader. Firm 1 knows (common knowledge of rationality) that firm 2 will follow (8.1), so it substitutes it in:

$$\max_{q_1} \Big(100 - q_1 - \frac{90 - q_1}{2}\Big)q_1 - 10q_1 \qquad (8.2)$$

Simplify the price: $100 - q_1 - 45 + \frac{q_1}{2} = 55 - \frac{q_1}{2}$, so the objective is $(55 - \frac{q_1}{2})q_1 - 10q_1 = 45q_1 - \frac{q_1^2}{2}$. Its derivative $45 - q_1 = 0$ gives $q_1 = 45$. (Tadelis writes the condition unsimplified: $100 - 2q_1 - 45 + q_1 - 10 = 0$.)

Step 3: the numbers.

	Stackelberg	Cournot	Change
q_1	45	30	$+15$
q_2	$\frac{90-45}{2} = 22.5$	30	-7.5
Q	67.5	60	$+7.5$ $(= \frac{9}{8} \times 60)$
p	$100 - 67.5 = 32.5$	40	-7.5
π_1	$(32.5 - 10) \times 45 =$ **1012.5**	900	$+$**112.5**
π_2	$(32.5 - 10) \times 22.5 =$ **506.25**	900	-393.75
$\pi_1 + \pi_2$	1518.75	1800	-281.25
consumer surplus $Q^2/2$	2278.125	1800	$+478.125$

> **"When two firms are competing by setting quantities, if one firm can somehow commit to move first, it will enjoy a first-mover advantage"** (p. 161).

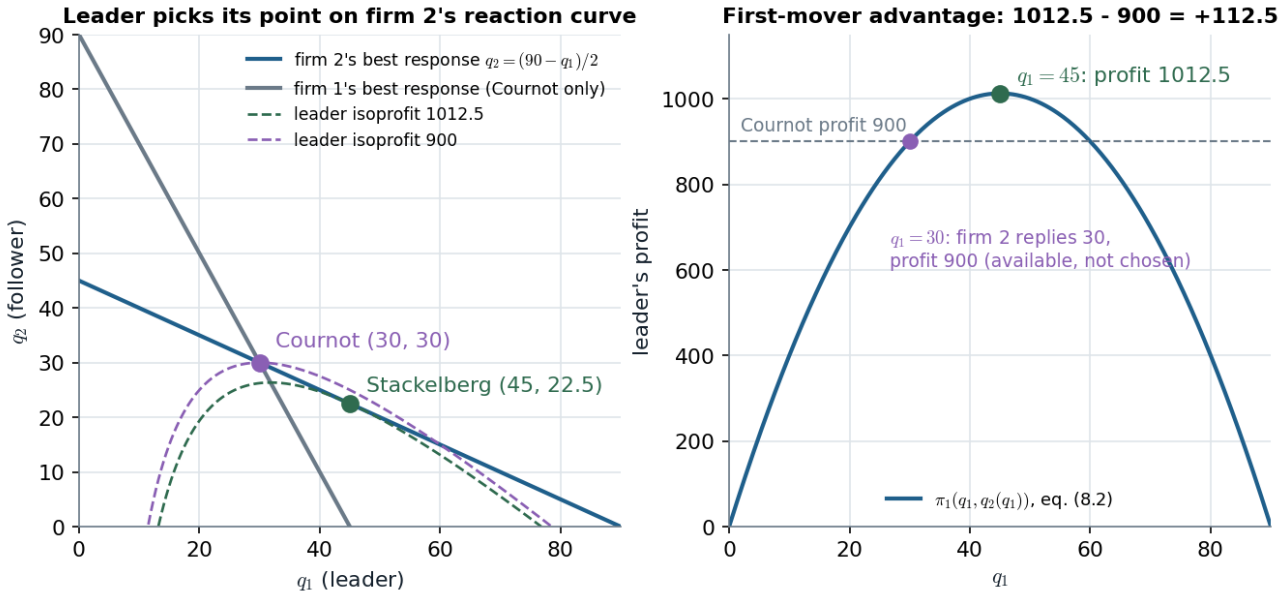


Figure — the leader chooses its point on firm 2's reaction curve. Left: quantity space with the Cournot and Stackelberg points and the leader's isoprofits. Right: the reduced objective (8.2). Source: Tadelis eqs. (8.1)–(8.2), numbers p. 161.

On the left, firm 1's isoprofit curves are hump-shaped and lower curves mean higher profit; the Stackelberg point is where the lowest isoprofit firm 1 can reach just touches firm 2's line. The Cournot point also lies on firm 2's line, so it is on the menu; on the right it sits on the same hump at 900, below the peak.

2.1 · Why the leader must gain, the follower must lose, and the total must fall

Gibbons's argument (p. 63) needs no algebra:

1. **Leader gains.** In the Stackelberg game firm 1 could choose its Cournot quantity, and firm 2 would reply with its Cournot quantity (because $q_2(30) = 30$). It gets 900 that way and chooses something else, so it gets at least 900.
2. **Price falls.** Aggregate output rises from $\frac{2}{3}(a - c)$ to $\frac{3}{4}(a - c)$.

3. **Industry profit falls.** Below the monopoly output, more output means less joint profit.
4. **Follower loses.** Joint profit fell and the leader's share rose, so the follower's fell.

2.2 · The general linear case (Gibbons pp. 62–63)

With $P(Q) = a - Q$ and marginal cost c: $R_2(q_1) = \frac{a-q_1-c}{2}$ (for $q_1 < a - c$), and the leader solves $\max_{q_1} q_1[a - q_1 - R_2(q_1) - c] = \max_{q_1} q_1 \frac{a-q_1-c}{2}$, so

$$q_1^* = \frac{a-c}{2}, \qquad R_2(q_1^*) = \frac{a-c}{4}, \qquad Q^S = \frac{3}{4}(a-c) > Q^C = \frac{2}{3}(a-c).$$

Check at $a = 100$, $c = 10$: 45, 22.5, 67.5 and 60. ✓

With slope b and costs c_1, c_2 (the companion's model, same steps): $q_1^S = \frac{a-2c_1+c_2}{2b}$, $q_2^S = \frac{a+2c_1-3c_2}{4b}$, and the follower is shut out when $c_2 \geq \frac{a+2c_1}{3}$; Cournot gives $q_i^C = \frac{a-2c_i+c_j}{3b}$.

> **Interactive companion.** The First-Mover Desk (`companions/companion-stackelberg-vs-cournot.html`) — move a, b, c_1, c_2 and read the first-mover advantage live; raise c_2 past $(a + 2c_1)/3$ (40 at the defaults) to watch the leader's commitment shut the follower out while Cournot still lets it in, and past $(a + c_1)/2$ (55) where both games become monopoly and the advantage is 0.

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3 · C — an equilibrium is not an outcome (p. 161)

> **"Writing down $(q_1, q_2) = (45, 22.5)$ is not identifying a Nash equilibrium because $q_1 = 45$ is not a best response to $q_2 = 22.5$."** (p. 161)

Check it: if firm 2 produced 22.5 regardless, firm 1's best response would be $q_1(22.5) = \frac{90-22.5}{2} = 33.75 \neq 45$. The pair of numbers is the **outcome**; the equilibrium is

$$\Big(q_1 = 45, \quad q_2(q_1) = \frac{90 - q_1}{2}\Big),$$

a number for the leader and **a function** for the follower, one action at each of its continuum of information sets. This is the Gibbons distinction of L03-05-subgame-perfection §5: outcome $(a_1^*, R_2(a_1^*))$, equilibrium $(a_1^*, R_2(\cdot))$.

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4 · C — Nash equilibria that are not subgame perfect (pp. 162–163)

Cournot's outcome is a Nash equilibrium of the Stackelberg game. Take $q_1 = 30$ and the *constant* strategy $q_2(q_1) = 30$ for every q_1.

Check	Computation	Holds?
firm 2's plan is a best reply on the path	$q_2(30)$ should be $\frac{90-30}{2} = 30$	yes
firm 1's $q_1 = 30$ is a best reply to "30 whatever you do"	$\max_{q_1}(100 - q_1 - 30 - 10)q_1 \Rightarrow q_1 = 30$	yes

Check	Computation	Holds?
firm 2's plan is a best reply **off** the path	at $q_1 = 45$ it plays 30, but $\frac{90-45}{2} = 22.5$	**no**

So it is a Nash equilibrium and not subgame perfect: "mutual best responses on the equilibrium path is all that Nash equilibrium requires" (p. 163). The only sequentially rational strategy for firm 2 is $q_2(q_1) = \frac{90-q_1}{2}$, which makes $(45, \frac{90-q_1}{2})$ **the unique SPE**.

A continuum of Nash equilibria (Remark, p. 163). Firm 2 threatens to flood the market unless firm 1 produces exactly 20:

$$q_2(q_1) = \begin{cases} 35 & \text{if } q_1 = 20 \\ 100 & \text{if } q_1 \neq 20 \end{cases}$$

Against it, $q_1 = 20$ earns $(100 - 20 - 35 - 10) \times 20 = 700 > 0$, while any other q_1 faces a price of $\max\{0, 100 - q_1 - 100\} = 0$ and earns $-10q_1 \leq 0$. So $q_1 = 20$ is a best reply, and 35 is firm 2's best reply to 20 since $\frac{90-20}{2} = 35$. Replacing 20 by any $\hat{q}_1 \in [0, 90]$ and 35 by $\frac{90-\hat{q}_1}{2}$ gives a Nash equilibrium for each $\hat{q}_1$: **every point of firm 2's reaction curve is a Nash outcome; one is subgame perfect.** The flooding threat is the textbook non-credible threat: having seen $q_1 \neq 20$, producing 100 would drive the price to 0 and lose firm 2 money.

> *Source detail.* The printed text on p. 163 says "$q_2 = 40$ is a best response to $q_1 = 20$"; with (8.1) the best response is $\frac{90-20}{2} = 35$, which is also the value in the displayed strategy. These notes use 35 (checked in L03-11-python §3).

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5 · U — more information can hurt (Gibbons p. 63)

In single-person decisions more information never hurts. Here firm 2 knows q_1 **and firm 1 knows that firm 2 knows it**, and firm 2 ends up worse off than in Cournot. Gibbons's check: let firm 1 move first but firm 2 *not observe* q_1. If firm 2 believed $q_1 = \frac{a-c}{2}$, it would reply $\frac{a-c}{4}$; but firm 1's best response to $\frac{a-c}{4}$ is $\frac{a-c-(a-c)/4}{2} = \frac{3}{8}(a-c)$, not $\frac{a-c}{2}$, so that belief is inconsistent. The unique Nash equilibrium of the unobserved game is Cournot, $\frac{a-c}{3}$ each. **Moving first is worth nothing unless the move is observed**, which is the L03-01-extensive-form lesson (information, not clock time) with a price tag.

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6 · Organizational reading

The Stackelberg leader is a firm whose capacity, plant or contract is sunk before rivals choose: a commitment in the sense of L03-06-credibility-entry-commitment §3.2, implemented by moving first rather than by a bond. Gibbons's example is General Motors' leadership role in the U.S. car industry (p. 61). The same structure, a leader that sets a variable and a follower that best-responds, is Leontief's monopoly union (Gibbons §2.1.C, pp. 64–68): the union sets the wage, the firm picks employment on its labour demand curve, and the outcome is inefficient, which Lecture 4 takes up as bargaining.

Links

- Cournot's best responses and Nash equilibrium: Lecture 2; IESDS in Cournot: Lecture 1.
- Collusion among Cournot firms when the game repeats: Lecture 6.
- Python: `stackelberg`, `cournot`, `flood`, L03-11-python §3. Figure code: `figures/make_figures.py::stackel`
- Exercises: Tadelis 8.4 (p. 170), 8.6 (p. 170–171), 8.9 (p. 172); Muñoz-Garcia & Toro-González Ch. 4, Exercise 11 (p. 180), 12 (p. 185), 13 (p. 189), 14 (p. 192), 15 (p. 195).

8.3.4 · Time-inconsistent preferences: a game against your future self

Claim. With exponential discounting, a plan made today is the plan you will want to follow tomorrow. With present bias (hyperbolic, $\beta\delta$ discounting) it is not, and a **sophisticated** decision maker treats her future self as another player and solves the problem by **backward induction**: a subgame-perfect equilibrium of a game with one person in it. This note also corrects a calculus slip in the book's worked example: with log utility the first-period choice is $K/2$, not $2K/3$. Prev: L03-07-stackelberg · Next: L03-09-formulas · Map: L03-00-map

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1 · C — exponential discounting is time consistent (pp. 166–167)

A stream $x_1, \dots, x_T$ is worth, in period 1,

$$v(x_1, \dots, x_T) = u(x_1) + \delta u(x_2) + \cdots + \delta^{T-1} u(x_T) = \sum_{t=1}^{T} \delta^{t-1} u(x_t) \qquad (8.3)$$

With $u = \ln$, budget K over three periods and $x_1 = K - x_2 - x_3$:

$$\max_{x_2, x_3} \ \ln(K - x_2 - x_3) + \delta \ln x_2 + \delta^2 \ln x_3.$$

First-order conditions: $-\frac{1}{K - x_2 - x_3} + \frac{\delta}{x_2} = 0$ and $-\frac{1}{K - x_2 - x_3} + \frac{\delta^2}{x_3} = 0$, so $x_2 = \delta x_1$ and $x_3 = \delta^2 x_1$. With $x_1 + x_2 + x_3 = K$:

$$x_1 = \frac{K}{1 + \delta + \delta^2}, \qquad x_2 = \frac{\delta K}{1 + \delta + \delta^2}, \qquad x_3 = \frac{\delta^2 K}{1 + \delta + \delta^2} \qquad (8.4)\text{–}(8.6)$$

Will period-2 self follow the plan? After consuming x_1 she has $K_2 = K - x_1 = \frac{(\delta + \delta^2)K}{1 + \delta + \delta^2}$ (8.7) and solves $\max_{x_2} \ln x_2 + \delta \ln(K_2 - x_2)$, so $\frac{1}{x_2} = \frac{\delta}{K_2 - x_2}$ and $x_2 = \frac{K_2}{1 + \delta}$ (8.8). Substituting (8.7):

$$x_2 = \frac{(\delta + \delta^2)K}{1 + \delta + \delta^2} \cdot \frac{1}{1 + \delta} = \frac{\delta K}{1 + \delta + \delta^2},$$

the same as (8.5). Exponential discounting is **time consistent** (p. 167).

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2 · C — present bias breaks it (pp. 167–168)

Hyperbolic ($\beta\delta$) discounting: the present is weighted 1 and *every* future period gets an extra factor $\beta \in (0, 1)$:

$$v(x_1, x_2, x_3) = \ln x_1 + \beta\delta \ln x_2 + \beta\delta^2 \ln x_3 \qquad (8.9)$$

Between today and tomorrow the discount is $\beta\delta$; between two future dates it is δ. Period-1 self wants to treat periods 2 and 3 alike (relative weight δ); when period 2 arrives, period-2 self discounts period 3 by $\beta\delta$ and wants more for herself. **Plans get revised.**

A sophisticated player 1 knows this and plays a game against "player 2", her period-2 self. Take $\delta = 1$, $\beta = \frac{1}{2}$.

2.1 · Backward induction: period-2 self (p. 168)

With budget K_2:

$$\max_{x_2} \ln x_2 + \tfrac{1}{2}\ln(K_2 - x_2), \qquad \frac{1}{x_2} - \frac{1/2}{K_2 - x_2} = 0 \;\Rightarrow\; K_2 - x_2 = \tfrac{1}{2}x_2 \;\Rightarrow\; x_2(K_2) = \tfrac{2}{3}K_2, \quad x_3(K_2) = \tfrac{1}{3}K_2.$$

This is player 2's strategy: a reply for each of the continuum of subgames indexed by K_2.

2.2 · Period-1 self (p. 169), and a correction

Player 1 substitutes the reply, with $K_2 = K - x_1$:

$$v_1(x_1) = \ln x_1 + \tfrac{1}{2}\ln\Big(\tfrac{2}{3}(K - x_1)\Big) + \tfrac{1}{2}\ln\Big(\tfrac{1}{3}(K - x_1)\Big).$$

The constants $\frac{2}{3}$ and $\frac{1}{3}$ sit inside logarithms, so they drop out of the derivative: $\frac{d}{dx_1}\ln\big(c(K - x_1)\big) = -\frac{1}{K - x_1}$ for any $c > 0$. Hence

$$v_1'(x_1) = \frac{1}{x_1} - \frac{1/2}{K - x_1} - \frac{1/2}{K - x_1} = \frac{1}{x_1} - \frac{1}{K - x_1} = 0 \quad \Rightarrow \quad \boxed{x_1 = \tfrac{K}{2}}, \quad x_2 = \tfrac{2}{3}\cdot\tfrac{K}{2} = \tfrac{K}{3}, \quad x_3 = \tfrac{K}{6}.$$

> *Source detail.* The book prints the first-order condition as $\frac{1}{x_1} - \frac{2}{3}\times\frac{0.5}{K - x_1} - \frac{1}{3}\times\frac{0.5}{K - x_1} = 0$, multiplying each term by the constant inside the logarithm, and concludes $x_1 = \frac{2}{3}K$ (p. 169). That is not the derivative of the objective printed one line above it. A grid search of v_1 (L03-11-python §4) peaks at $x_1 = 0.5K$, and $v_1(\frac{K}{2}) > v_1(\frac{2K}{3})$. If you are examined on this example, state the objective and differentiate it; do not reproduce the printed number.

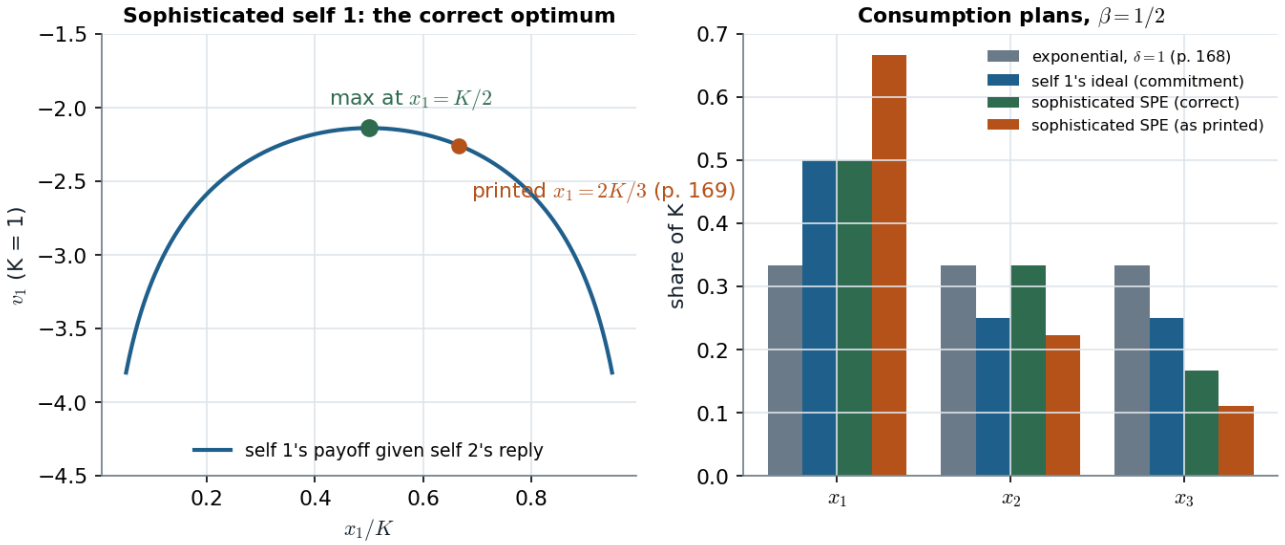


Figure — the sophisticated three-period problem with $\beta = \frac{1}{2}$, $\delta = 1$, $u = \ln$. Left: self 1's objective given

self 2's reply. Right: four consumption plans. Source: Tadelis §8.3.4, eqs. (8.4)–(8.9), pp. 166–169; corrected optimum computed here.

The right panel separates the two effects. Self 1's *own* ideal plan (what she would lock in if she could) is $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$: present bias alone already moves consumption to period 1. The sophisticated equilibrium keeps $x_1 = \frac{1}{2}$ but splits the rest $\frac{2}{3} : \frac{1}{3}$ instead of $\frac{1}{2} : \frac{1}{2}$, because period-2 self grabs more than self 1 wants her to.

2.3 · What survives of the book's story

Claim on p. 169	With the correct derivative
Player 2 overconsumes relative to player 1's wishes	**yes**: $x_2 = \frac{K}{3}$ against the $\frac{K}{4}$ player 1 would choose for her
Player 1 overconsumes to leave player 2 less	**not with log utility**: $x_1 = \frac{K}{2}$ either way. With ln, the ratio x_2/x_3 chosen by self 2 does not change self 1's marginal value of the remaining budget, so self 1 has no reason to shrink it
The result differs from the exponential benchmark ($\frac{K}{3}$ each)	**yes**, but because of β, not because of the game

> **The structure is the lesson, not the number**: a time-inconsistent decision maker who anticipates her future choices is solving for a subgame-perfect equilibrium, and **she would pay to commit** (exercise 8.16 asks how much; not solved here). Naive players (who do not anticipate) and sophisticated ones (who do) are distinguished in the behavioural literature: Laibson (1997) on savings, O'Donoghue and Rabin (1999) on procrastination (p. 169).

Links

- Previous: commitment as a source of value, L03-06-credibility-entry-commitment §3.
- Discounting in repeated games: Lecture 5.
- Python: `v1` and the commitment plan, L03-11-python §4; figure code `figures/make_figures.py::hyperbolic`.
- Exercises: Tadelis 8.14, 8.15, 8.16 (p. 174).

Lecture 3 · Formulas — every definition, result and number, the exam specification mapped

Prev: L03-08-time-inconsistency · Next: L03-10-pitfalls · Map: L03-00-map

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0 · The exam specification, mapped

The lecture has no slides yet, so the specification is the study guide's block for Lecture 3 (`Map/study-guide.md`: Tadelis Ch. 7–8, Gibbons §2.1.A, §2.2.A, §2.4; formal MWG §7.C, §9.A–9.B; applications Cabral §7.2) read against the course's learning goal "solve games of every major type, choosing the appropriate tools according to when each agent moves and what each agent knows" (Lecture 1 slides, p. 7). Item by item:

#	You should be able to	Note
1	Draw a game tree from a verbal description: players, order, actions, information sets, Nature, payoffs	L03-01-extensive-form §2–4
2	Say whether a game has perfect or imperfect information, and why every simultaneous game is imperfect	L03-01-extensive-form §5
3	Write each player's pure-strategy set and count it, $\lvert S_i\rvert = \prod m_k$	L03-02-strategies §1
4	Distinguish mixed from behavioral strategies; state Kuhn's equivalence under perfect recall	L03-02-strategies §2–4
5	Build the normal form of a tree and find all pure Nash equilibria	L03-03-normal-form-and-paths §1–2
6	Identify on- and off-path information sets and the threat that supports a Nash equilibrium	L03-03-normal-form-and-paths §3
7	Solve a finite perfect-information game by backward induction; state Zermelo / Prop. 8.1 and when the solution is unique	L03-04-sequential-rationality-backward-induction
8	List the proper subgames of a tree and justify each	L03-05-subgame-perfection §2
9	Find all pure SPE, branching when a subgame has several equilibria (voluntary BoS, MAD)	L03-05-subgame-perfection §4
10	Write an SPE as complete strategies, not as an outcome	L03-05-subgame-perfection §5; L03-07-stackelberg §3

#	You should be able to	Note
11	Explain why a threat is or is not credible, and value a commitment	L03-06-credibility-entry-commitment
12	Solve Stackelberg, compare with Cournot, and exhibit a Nash equilibrium that is not subgame perfect	L03-07-stackelberg
13	Treat a time-inconsistent decision maker as a game against her future self	L03-08-time-inconsistency

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1 · Definitions and results

Ref.	Statement	Derived in
Elements 1–7 (pp. 130–131)	players; payoffs; order of moves; actions when moving; **knowledge when moving**; Nature's distributions; common knowledge of 1–6	L03-01-extensive-form §2
Def. 7.1 (p. 133)	game tree: nodes, precedence (transitive, asymmetric, incomplete), one predecessor per node, root x_0, terminal nodes Z, $v_i : Z \to \mathbb{R}$	L03-01-extensive-form §3
Def. 7.2 (p. 135)	information sets $h_i \in H_i$ partition i's nodes; same set ⇒ cannot tell apart ⇒ same actions	§4
Def. 7.3 (p. 136)	perfect information ⇔ all information sets singletons and no Nature	§5
Def. 7.4 (p. 139)	pure strategy $s_i : H_i \to A_i$, $s_i(h_i) \in A_i(h_i)$	L03-02-strategies §1
Count (p. 139)	$\lvert S_i \rvert = m_1 \times \cdots \times m_k$	§1.1
Def. 7.5 (p. 139)	mixed strategy: distribution over S_i	§2
Def. 7.6 (p. 140)	behavioral strategy: independent distribution over $A_i(h_i)$ at each h_i	§3
Mixed → behavioral (p. 141)	$\sigma_i(a \mid h) = \sum_{s_i : s_i(h)=a} p_{s_i}$	§3.1
Def. 7.7 (p. 142)	perfect recall; **Kuhn (1953)**: mixed ≡ behavioral under perfect recall	§4
Normal form of a tree (pp. 143–144)	unique; the converse fails (Fig. 7.12)	L03-03-normal-form-and-paths §1

Ref.	Statement	Derived in
Def. 7.8 (p. 146)	information set on / off the equilibrium path: reached with positive / zero probability	§3
Def. 8.1 (p. 152)	sequential rationality: best response at **each** information set	L03-04-sequential-rationality-backward-induction §2
Prop. 8.1 (Zermelo, p. 153)	finite + perfect information ⇒ a sequentially rational backward-induction solution; unique if no player has equal payoffs at two terminal nodes	§2
Cor. 8.1 (p. 153)	⇒ a sequentially rational pure-strategy Nash equilibrium exists; unique under the same condition	§2
MWG **Prop. 9.B.1**	Zermelo: finite perfect-information games have a pure NE derivable by backward induction	§2
Def. 8.2 (pp. 154–155)	proper subgame: one node, all successors, no broken information sets	L03-05-subgame-perfection §2
Def. 8.3 (p. 157)	SPE: restriction of σ^* is a Nash equilibrium in **every** proper subgame	§3
Fact (p. 158)	finite perfect information: SPE = profiles surviving backward induction	§3
MWG **Def. 9.B.1–9.B.2**, **Prop. 9.B.2–9.B.3**	subgame; SPNE; existence and uniqueness under perfect information; replace-a-subgame reduction	§2–4 (G)
Gibbons outcome vs equilibrium (pp. 125–126)	outcome $(a_1^*, R_2(a_1^*))$; SPE $(a_1^*, R_2(\cdot))$	§5
(8.1) (p. 160)	$q_2(q_1) = \dfrac{90 - q_1}{2}$	L03-07-stackelberg §2
(8.2) (p. 161)	$\max_{q_1}(100 - q_1 - \frac{90-q_1}{2})q_1 - 10q_1$	§2
Gibbons general (pp. 62–63)	$q_1^* = \frac{a-c}{2}$, $q_2^* = \frac{a-c}{4}$, $Q^S = \frac{3}{4}(a-c) > Q^C = \frac{2}{3}(a-c)$	§2.2
Asymmetric, slope b	$q_1^S = \frac{a-2c_1+c_2}{2b}$, $q_2^S = \frac{a+2c_1-3c_2}{4b}$; follower out iff $c_2 \geq \frac{a+2c_1}{3}$; Cournot $q_i^C = \frac{a-2c_i+c_j}{3b}$	§2.2
Credibility threshold	bond B makes retaliation credible iff $F_I > D_I - B$, i.e. $B > B^* = D_I - F_I$	L03-06-credibility-entry-commitment §3.1
(8.3) (p. 166)	$v = \sum_t \delta^{t-1} u(x_t)$	L03-08-time-inconsistency §1
(8.4)–(8.6) (p. 167)	$x_t = \dfrac{\delta^{t-1} K}{1 + \delta + \delta^2}$ for $u = \ln$: time consistent	§1

Ref.	Statement	Derived in
(8.9) (p. 168)	$\ln x_1 + \beta\delta \ln x_2 + \beta\delta^2 \ln x_3$	§2

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2 · Every number and worked case

2.1 · Sequential Battle of the Sexes (Figs. 7.7, 8.6; pp. 138–158)

$S_1 = \{O, F\}$, $S_2 = \{oo, of, fo, ff\}$. Normal form rows O: $(2,1), (2,1), (0,0), (0,0)$; F: $(0,0), (1,2), (0,0), (1,2)$. **NE** $\{(O, oo), (O, of), (F, ff)\}$; **SPE** (O, of), outcome $(2,1)$. Simultaneous version: NE (O, o), (F, f) and the mixed one with O at $\frac{2}{3}$, o at $\frac{1}{3}$ (p. 145).

2.2 · Behavioral ↔ mixed (pp. 140–142)

$\sigma_2(o \mid O) = \frac{1}{3}$, $\sigma_2(o \mid F) = \frac{1}{2} \Leftarrow (\frac{1}{3}, 0, \frac{1}{6}, \frac{1}{2})$ or $(\frac{1}{6}, \frac{1}{6}, \frac{1}{3}, \frac{1}{3})$; general $(t, \frac{1}{3} - t, \frac{1}{2} - t, \frac{1}{6} + t)$, $t \in [0, \frac{1}{3}]$.

2.3 · Absent-minded driver (p. 142)

$V(p) = -3p^2 + 2p + 1$, $p^* = \frac{1}{3}$, $V^* = \frac{4}{3}$.

2.4 · Gibbons Fig. 2.4.1 and the three-move game (pp. 59–60, 116–128)

NE $\{(R, (R', L')), (L, (R', R'))\}$; SPE $(R, (R', L'))$, outcome $(2,1)$. Three-move game: backward induction L'', L', L; outcome $(2,0)$.

2.5 · Centipede (Fig. 8.8, p. 159)

Stops $(1,1), (0,3), (2,2), (1,4)$; end $(3,3)$. Unique backward-induction outcome $(1,1)$.

2.6 · Voluntary BoS (Figs. 8.2–8.7, pp. 154–159)

Two proper subgames. NE $\{(YO, o), (NO, f), (NF, f)\}$; SPE $\{(YO, o), (NF, f)\}$.

2.7 · Card game (Fig. 8.5, p. 156)

Only proper subgame: the whole game. (Fig. 7.6 variant: calling is worth 0 in expectation, folding -1.)

2.8 · Mutually assured destruction (Fig. 8.11, pp. 163–166)

War stage NE: (R, r) with $(-5, -5)$, (D, d) with $(-100, -100)$. Six pure NE; **SPE** $\{(IR, Nr), (ED, Bd)\}$.

2.9 · Entry and commitment (Cabral pp. 168–170; MWG pp. 268–269)

Cabral: out $(0, 50)$, accommodate $(10, 20)$, retaliate $(-10, 10)$ [text: incumbent -10]. NE $\{(e, \bar{r}), (\bar{e}, r)\}$; SPE $(e, \bar{r})$. Bond 40: accommodation $\to -20$, retaliate, entrant out, incumbent 50; **value of commitment 30**; $B^* = 10$ (30 with the text's value). MWG 9.B.1: out $(0, 2)$, fight $(-3, -1)$, accommodate $(2, 1)$; SPE (in, accommodate).

2.10 · Stackelberg vs Cournot (pp. 160–163)

	q_1	q_2	p	π_1	π_2
Stackelberg	45	22.5	32.5	1012.5	506.25
Cournot	30	30	40	900	900

First-mover advantage +112.5; follower −393.75; $Q^S/Q^C = \frac{9}{8}$. Nash, not SPE: (30, $q_2 \equiv 30$); the flood strategy with $\widehat{q}_1 = 20$, on-path reply 35 (printed "40" on p. 163). Unobserved q_1: Cournot again (Gibbons p. 63–64).

2.11 · Hyperbolic discounting (pp. 166–169)

$\delta = 1$, $\beta = \frac{1}{2}$: self 2 $x_2 = \frac{2}{3}K_2$, $x_3 = \frac{1}{3}K_2$. Self 1 (corrected) $x_1 = \frac{K}{2}$, so $(\frac{1}{2}, \frac{1}{3}, \frac{1}{6})K$; printed $\frac{2}{3}K$. Self 1's committed ideal $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})K$. Exponential benchmark $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})K$.

Lecture 3 · Pitfalls and an original True/False self-test

Prev: L03-09-formulas · Next: L03-11-python · Map: L03-00-map

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Part I — the confusions this material produces

1 · An outcome is not an equilibrium

Error: answering "the SPE is $(45, 22.5)$" or "the SPE is (O, o)". **Minimal counterexample:** in Stackelberg, $q_1 = 45$ is not a best response to a constant $q_2 = 22.5$ (the best response is 33.75), so the pair is not even a Nash equilibrium of anything. The SPE is $(45,\ q_2(q_1) = \frac{90-q_1}{2})$; in the BoS it is (O, of). → L03-07-stackelberg §3, L03-05-subgame-perfection §5.

2 · A strategy is not an action

Error: giving player 2 in the sequential BoS the strategy set $\{o, f\}$. **Counterexample:** she has two information sets, so $S_2 = \{oo, of, fo, ff\}$. The plan must name an action at **every** information set, including ones the plan makes unreachable (NO vs NF). → L03-02-strategies §1.

3 · "Sequential" is about information, not time

Error: treating a later mover as a follower even when she cannot see the earlier move. **Counterexample:** Gibbons's unobserved Stackelberg game: firm 1 moves first, firm 2 does not see q_1, and the unique equilibrium is Cournot. → L03-07-stackelberg §5, L03-01-extensive-form §2.

4 · Complete ≠ perfect information

Error: "complete information, so backward induction applies." **Counterexample:** the voluntary BoS has complete information and a non-singleton information set; backward induction cannot start, subgame perfection can. → L03-05-subgame-perfection §1.

5 · Every Nash equilibrium of a tree is "reasonable"

Error: reporting (F, ff) or $(\bar{e}, r)$ as predictions. **Counterexample:** they rest on actions that are not optimal at the node where they would be taken. Nash equilibrium never tests those nodes; subgame perfection does. → L03-03-normal-form-and-paths §3.

6 · Subgame perfection always gives a unique answer

Error: "SPE eliminates multiplicity." **Counterexample:** MAD has two pure SPE, because the war-stage subgame has two Nash equilibria; the voluntary BoS also has two. Uniqueness needs perfect information **and** no payoff ties (Prop. 8.1). → L03-05-subgame-perfection §4.

7 · Any node with a singleton information set starts a subgame

Error: listing the card game's x_1 as a subgame root. **Counterexample:** the subgame at x_1 would contain x_3 and hence x_4, which x_1 does not precede: it cuts an information set. → L03-05-subgame-perfection §2.

8 · Committing is free / moving first always helps

Error: "the first mover always wins" or "a bond can only cost the incumbent." **Counterexamples:** in Matching Pennies the first mover loses (p. 130); a bond below $B^* = D_I - F_I$ lowers the incumbent's payoff and changes nothing; a bond above it raises the incumbent's equilibrium payoff by lowering one of its own terminal payoffs. → L03-06-credibility-entry-commitment §3.

9 · Backward induction always finds the efficient outcome

Error: expecting rational players to reach the high-payoff end of a game. **Counterexample:** Centipede: $(1, 1)$ instead of $(3, 3)$; trust game: $(0, 0)$ instead of $(1, 1)$. → L03-04-sequential-rationality-backward-induction §4.

10 · Copying the book's slips

Two printed numbers in Ch. 8 do not follow from the book's own equations: the on-path reply in the flooding remark is $\frac{90-20}{2} = 35$, not 40 (p. 163); and the sophisticated first-period choice in §8.3.4 is $x_1 = \frac{K}{2}$, not $\frac{2}{3}K$ (p. 169). → L03-07-stackelberg §4, L03-08-time-inconsistency §2.2.

Part II — True/False self-test (original items)

Scoring as in the exam: $\max\{0,\ \text{correct} - \text{wrong}/2\} \times$ item value; blanks do not count as wrong (rules/00_estilo_avaliacao.md).

1. A game in which player 2 moves one hour after player 1 but without observing player 1's choice is strategically equivalent to the simultaneous-move game with the same payoffs. **True.** What defines a sequential game is what the later mover knows (element 5, p. 131); with one information set containing all of player 1's choices, the tree is the simultaneous game's tree.

2. A game of complete information is necessarily a game of perfect information. **False.** Complete information means payoffs are common knowledge; perfect information also requires singleton information sets and no Nature (Def. 7.3). Any simultaneous game is a counterexample.

3. A player with three information sets, having 2, 2 and 3 actions respectively, has 7 pure strategies. **False.** Strategies multiply, they do not add: $2 \times 2 \times 3 = 12$.

4. In the sequential Battle of the Sexes, the strategies oo and of of player 2 always produce the same outcome. **False.** They coincide only when player 1 plays O; after F, oo leads to $(0, 0)$ and of to $(1, 2)$.

5. Every mixed strategy in an extensive-form game can be replicated by a behavioral strategy, and in a game of perfect recall the converse also holds. **True.** Mixed → behavioral by conditioning (p. 141); behavioral → mixed under perfect recall is Kuhn's theorem (p. 142).

6. The behavioral strategy that replicates a given mixed strategy may correspond to several mixed strategies, but the mixed strategy that replicates a given behavioral strategy is unique. **False.** It is the other way round: many mixed strategies map to the same behavioral one (Fig. 7.9's example has a one-parameter family); conditioning a mixed strategy gives exactly one behavioral strategy.

7. In a Nash equilibrium of an extensive-form game, a player's action at an information set that is off the equilibrium path does not affect her own equilibrium payoff. **True.** Off-path sets are reached with probability 0 given the profile, so her payoff is the same whatever she plans there; that is why Nash cannot discipline those plans.

8. In a Nash equilibrium, a player's action at an off-path information set cannot affect the other players' choices. **False.** It can, and that is the whole problem: player 2's f at x_1 is what keeps player 1 at F in (F, ff).

9. Every subgame-perfect equilibrium is a Nash equilibrium. **True.** The whole game is a proper subgame, so the restriction to it, the profile itself, is a Nash equilibrium.

10. Every finite game of perfect information has a subgame-perfect equilibrium in pure strategies. **True.** Backward induction constructs one (Prop. 8.1, Cor. 8.1, and the Fact on p. 158).

11. If a finite game of perfect information has two terminal nodes giving player 1 the same payoff, it has more than one subgame-perfect equilibrium. **False.** Uniqueness may fail but need not: if the tie is never the deciding comparison at any node, the solution is still unique. The proposition gives a sufficient condition for uniqueness, not a necessary one.

12. In a game whose only proper subgame is the whole game, every Nash equilibrium is subgame perfect. **True.** Definition 8.3 then only requires a Nash equilibrium of the whole game (MWG Exercise 9.B.2(a)).

13. A node that is a singleton information set always starts a proper subgame. **False.** It must also not cut any later information set; the card game's x_1 and x_2 are singletons and start no subgame.

14. In the voluntary Battle of the Sexes, (NO, f) is not a Nash equilibrium because O is never played. **False.** It is a Nash equilibrium (player 1 gets 1.5 against f whatever she plans after Y); it fails subgame perfection because (O, f) is not an equilibrium of the subgame at x_1.

15. In the mutually assured destruction game the war stage is off the path in every subgame-perfect equilibrium, so the war-stage payoffs do not affect which equilibria are subgame perfect. **False.** The war stage is off the path in both SPE, but which of its equilibria is expected decides whether player 2 backs down and whether player 1 escalates.

16. In the entry game of Cabral Fig. 7.7, $(\bar{e}, r)$ is a Nash equilibrium. **True.** Against r the entrant prefers 0 to -10; against $\bar{e}$ the incumbent gets 50 whatever it plans, so r is a best reply. It is not subgame perfect.

17. A bond that lowers the incumbent's payoff from accommodating can never raise its equilibrium payoff. **False.** In Cabral's example it raises it from 20 to 50: the lower payoff sits at a node the bond makes unreachable, and it makes retaliation credible.

18. In Cabral's entry game, any bond $B > 0$ makes the incumbent better off. **False.** Only $B > B^* = 10$ changes the incumbent's post-entry choice; a smaller bond only lowers its accommodation payoff, which is then its equilibrium payoff.

19. In the Stackelberg game with $p = 100 - q$ and unit cost 10, the leader's profit exceeds its Cournot profit, the follower's is lower, and industry profit is lower than in Cournot. **True.** $1012.5 > 900$, $506.25 < 900$, $1518.75 < 1800$.

20. The pair $(q_1, q_2) = (45, 22.5)$ is a subgame-perfect equilibrium of the Stackelberg game. **False.** It is the SPE outcome. The equilibrium specifies firm 2's quantity after every q_1: $(45, \frac{90-q_1}{2})$.

21. The Cournot quantities $(30, 30)$ can be the outcome of a Nash equilibrium of the Stackelberg game. **True.** $q_1 = 30$ with $q_2(q_1) \equiv 30$ is a Nash equilibrium (p. 162); it is not subgame perfect because 30 is not firm 2's best reply to other q_1.

22. If firm 1 still moves first but firm 2 cannot observe q_1, firm 2 earns more than in the Stackelberg game. **True.** Without observation the unique Nash equilibrium is Cournot (Gibbons pp. 63–64), where

firm 2 earns 900 instead of 506.25: knowing q_1, and being known to know it, hurts firm 2.

23. In the Stackelberg game, total output is higher and price lower than in Cournot. **True.** $\frac{3}{4}(a-c) > \frac{2}{3}(a-c)$; $32.5 < 40$.

24. A player with exponential discounting and log utility revises her consumption plan when the second period arrives. **False.** Exponential discounting is time consistent: the period-2 choice (8.8) equals the planned (8.5).

25. A sophisticated hyperbolic discounter solves her consumption problem by backward induction against her future selves. **True.** She treats period-2 self as a player who best-responds to the remaining budget, and chooses period-1 consumption given that reply: an SPE (pp. 168–169).

26. Backward induction predicts that the Centipede game ends at the first node, and laboratory subjects typically behave this way. **False.** The first half is true; most lab subjects continue for some time (McKelvey and Palfrey, 1992), though chess players mostly stop at once (Palacios-Huerta and Volij, 2009; p. 160).

Links

- Map: L03-00-map. Formulas: L03-09-formulas. Python checks for the numerical items: L03-11-python.
- Quizzes can be generated from this note and L03-09-formulas with `/quiz-gen`.

Lecture 3 · Computation in Python

> Every routine the lecture needs, each followed by the `assert` checks that must pass against the book's own numbers. Python standard library only (no numpy needed here). The blocks run in order as one script; all of them were run together before this note was written. Map: L03-00-map · Formulas: L03-09-formulas · Pitfalls: L03-10-pitfalls

Reuse. The project's existing checkers (`Listas/lista-aulas-01-02-check.py`, `Aula/notes-aulas-01-02/check_part2.py`) cover normal-form games and bargaining shares; neither represents a game tree, so the tree solver below is new. Bargaining by backward induction is reused, not rebuilt, in Lecture 4 (`check_part2.py::proposer_share`).

Design. A tree is nested tuples: `T(payoffs)` for a terminal node, `D(player, info, moves)` for a decision node, `N(moves)` for Nature. Nodes that share an `info` label form one information set, which is how imperfect information is encoded. Everything else (strategies, the normal form, Nash equilibria, subgames, SPE) is computed from that one object by brute force, which is exact and fast for textbook-sized trees.

---

1 · Trees, strategies, Nash and subgame-perfect equilibria → L03-01-extensive-form to L03-05-subgame-perfection

1.1 · The solver

```
from itertools import product
from math import prod, isclose


def T(*payoffs):
    """Terminal node: one payoff per player (player 1 first)."""
    return ("T", tuple(payoffs))


def D(player, info, moves):
    """Decision node of `player` (1-based) in information set `info`; moves: {action:
     ↪  child}."""
    return ("D", player, info, dict(moves))


def N(moves):
    """Nature: {label: (probability, child)}."""
    return ("N", dict(moves))


def nodes(tree, path=()):
    """Yield (path, node) for every node; a path is the tuple of actions from the
     ↪  root."""
    yield path, tree
    if tree[0] == "D":
        for a, child in tree[3].items():
            yield from nodes(child, path + (a,))
```

```
    elif tree[0] == "N":
        for a, (_, child) in tree[1].items():
            yield from nodes(child, path + (a,))


def infosets(tree):
    """{info: (player, actions, [member paths])}. Raises if Definition 7.2(3)
    ↪ fails."""
    H = {}
    for path, x in nodes(tree):
        if x[0] == "D":
            _, i, h, moves = x
            if h in H:
                assert H[h][0] == i and H[h][1] == tuple(moves), f"info set {h}: Def.
                ↪ 7.2(3) fails"
                H[h][2].append(path)
            else:
                H[h] = (i, tuple(moves), [path])
    return H


def players(tree):
    return sorted({v[0] for v in infosets(tree).values()})


def pure_strategies(tree, i):
    """Definition 7.4: every map H_i -> A_i(h_i), as a dict {info: action}."""
    H = {h: v for h, v in infosets(tree).items() if v[0] == i}
    hs = sorted(H)
    return [dict(zip(hs, acts)) for acts in product(*(H[h][1] for h in hs))]


def n_pure_strategies(tree, i):
    """|S_i| = m_1 x m_2 x ... x m_k (Tadelis p. 139)."""
    return prod(len(v[1]) for v in infosets(tree).values() if v[0] == i)


def payoff(node, profile):
    """Expected payoff vector from `node` when everyone follows `profile` ({info:
    ↪ action})."""
    if node[0] == "T":
        return node[1]
    if node[0] == "D":
        return payoff(node[3][profile[node[2]]], profile)
    out = None
    for p, child in node[1].values():
        v = [p * u for u in payoff(child, profile)]
        out = v if out is None else [a + b for a, b in zip(out, v)]
    return tuple(out)
```

```
def profiles(tree):
    """All pure-strategy profiles, as (tuple of per-player strategies, merged
     ↪ dict)."""
    P = players(tree)
    for combo in product(*(pure_strategies(tree, i) for i in P)):
        merged = {}
        for s in combo:
            merged.update(s)
        yield combo, merged


def _is_nash_from(tree, root, combo):
    """No player gains by a unilateral switch of pure strategy, payoffs measured from
     ↪ `root`."""
    P = players(tree)
    merged = {k: v for s in combo for k, v in s.items()}
    base = payoff(root, merged)
    for k, i in enumerate(P):
        for s_dev in pure_strategies(tree, i):
            u = payoff(root, {**merged, **s_dev})[i - 1]
            if u > base[i - 1] + 1e-12:
                return False
    return True


def pure_nash(tree):
    """Every pure-strategy Nash equilibrium of the normal form of `tree` (Section
     ↪ 7.3)."""
    return [combo for combo, _ in profiles(tree) if _is_nash_from(tree, tree, combo)]


def subgame_roots(tree):
    """Definition 8.2: a node that is a singleton information set (or the root) and
     ↪ whose
    successors' information sets lie entirely inside its subtree."""
    H = infosets(tree)
    where = {}
    for h, (_, _, members) in H.items():
        for m in members:
            where[m] = h
    roots = []
    for path, x in nodes(tree):
        if x[0] == "T":
            continue
        if x[0] == "D" and len(H[x[2]][2]) > 1:
            continue
        inside = all(all(m[:len(path)] == path for m in H[where[q]][2])
                     for q, y in nodes(x, path) if y[0] == "D")
        if inside:
```

```
            roots.append((path, x))
    return roots


def spe(tree):
    """Definition 8.3 by brute force: the pure profiles that are a Nash equilibrium
     ↪  in every
    proper subgame (the whole game included)."""
    roots = subgame_roots(tree)
    return [combo for combo, _ in profiles(tree)
            if all(_is_nash_from(tree, r, combo) for _, r in roots)]


def backward_induction(tree):
    """Proposition 8.1 for finite perfect-information games without Nature.
    Returns (payoff vector, plan {info: action}). Ties go to the first-listed
 ↪  action."""
    if tree[0] == "T":
        return tree[1], {}
    assert tree[0] == "D", "backward induction here needs perfect information, no
     ↪  Nature"
    _, i, h, moves = tree
    best, plan = None, {}
    for a, child in moves.items():
        v, sub = backward_induction(child)
        plan.update(sub)
        if best is None or v[i - 1] > best[1][i - 1]:
            best = (a, v)
    plan[h] = best[0]
    return best[1], plan


def label(combo):
    """Readable profile: each strategy as its actions in information-set order."""
    return tuple("".join(s[h] for h in sorted(s)) for s in combo)
```

spe checks Definition 8.3 literally: a profile passes if, from the root of every proper subgame, no player gains by switching to another of her pure strategies. Because payoffs are measured from the subgame's root, only the actions inside the subgame matter, which is exactly "the restriction is a Nash equilibrium of the subgame". It finds **pure-strategy** SPE only; the mixed ones of exercises 8.1–8.2 are outside its reach (ceiling stated; add support enumeration per subgame if ever needed).

1.2 · Perfect information: Battle of the Sexes, Gibbons, Centipede

```
# --- Sequential-move Battle of the Sexes, Fig. 7.7 / 8.6 (pp. 138, 157)
 ↪  -------------
bos_seq = D(1, "x0", {"O": D(2, "x1", {"o": T(2, 1), "f": T(0, 0)}),
                      "F": D(2, "x2", {"o": T(0, 0), "f": T(1, 2)})})
bos_sim = D(1, "x0", {"O": D(2, "h2", {"o": T(2, 1), "f": T(0, 0)}),
                      "F": D(2, "h2", {"o": T(0, 0), "f": T(1, 2)})})
assert n_pure_strategies(bos_seq, 2) == 4 and n_pure_strategies(bos_sim, 2) == 2
```

```
assert sorted(map(label, pure_nash(bos_seq))) == [("F", "ff"), ("O", "of"), ("O",
 ↪  "oo")]   # p. 145
assert list(map(label, spe(bos_seq))) == [("O", "of")]
 ↪  # p. 157
assert backward_induction(bos_seq) == ((2, 1), {"x1": "o", "x2": "f", "x0": "O"})
assert sorted(map(label, pure_nash(bos_sim))) == [("F", "f"), ("O", "o")]
assert len(subgame_roots(bos_sim)) == 1                    # simultaneous: only the
 ↪  whole game

# --- Gibbons Figure 2.4.1 (p. 116): NE (R,(R',L')) and (L,(R',R')); SPE only the
 ↪  first --
gib = D(1, "a", {"L": D(2, "b", {"L'": T(3, 1), "R'": T(1, 2)}),
                 "R": D(2, "c", {"L'": T(2, 1), "R'": T(0, 0)})})
assert sorted(map(label, pure_nash(gib))) == [("L", "R'R'"), ("R", "R'L'")]
assert list(map(label, spe(gib))) == [("R", "R'L'")]

# --- Gibbons three-move game (pp. 59-60): backward induction ends the game at once
 ↪  -----
gib3 = D(1, "a", {"L": T(2, 0),
                  "R": D(2, "b", {"L'": T(1, 1),
                                  "R'": D(1, "c", {"L''": T(3, 0), "R''": T(0,
                                   ↪  2)})})})
v, plan = backward_induction(gib3)
assert v == (2, 0) and plan == {"c": "L''", "b": "L'", "a": "L"}

# --- Centipede, Fig. 8.8 (p. 159): unique BI outcome (1,1), although (3,3) is
 ↪  available --
cent = D(1, "1a", {"N": T(1, 1), "C": D(2, "2a", {"n": T(0, 3), "c": D(1, "1b", {
    "N": T(2, 2), "C": D(2, "2b", {"n": T(1, 4), "c": T(3, 3)})})})})
v, plan = backward_induction(cent)
assert v == (1, 1) and plan == {"2b": "n", "1b": "N", "2a": "n", "1a": "N"}
assert len(spe(cent)) == 1
```

1.3 · Imperfect information and applications: voluntary BoS, card game, MAD, entry

```
# --- Voluntary Battle of the Sexes, Fig. 8.2/8.4 (pp. 154-158)
 ↪  --------------------------
vbos = D(1, "x0", {"Y": D(1, "x1", {"O": D(2, "h2", {"o": T(2, 1), "f": T(0, 0)}),
                                    "F": D(2, "h2", {"o": T(0, 0), "f": T(1, 2)})}),
                   "N": T(1.5, 1.5)})
lab = lambda c: (c[0]["x0"] + c[0]["x1"], c[1]["h2"])          # Tadelis writes YO,
 ↪  NF, ...
assert sorted(map(lab, pure_nash(vbos))) == [("NF", "f"), ("NO", "f"), ("YO", "o")]
 ↪  # p. 158
assert sorted(map(lab, spe(vbos))) == [("NF", "f"), ("YO", "o")]
assert [p for p, _ in subgame_roots(vbos)] == [(), ("Y",)]       # exactly two
 ↪  subgames

# --- Card game, Fig. 8.5 (p. 156): the only proper subgame is the whole game
 ↪  -----------
```

```
card = N({"A": (0.5, D(1, "x1", {"N": T(-1, 1), "Y": D(2, "h2", {"C": T(2, -2), "F":
 ↪  T(1, -1)})})),
           "K": (0.5, D(1, "x2", {"N": T(-1, 1), "Y": D(2, "h2", {"C": T(-2, 2), "F":
            ↪  T(1, -1)})})})})
assert [p for p, _ in subgame_roots(card)] == [()]

# --- Mutually assured destruction, Fig. 8.11 (pp. 163-166)
 ↪  -------------------------------
war = lambda: D(1, "w1", {"R": D(2, "w2", {"r": T(-5, -5), "d": T(-100, -100)}),
                          "D": D(2, "w2", {"r": T(-100, -100), "d": T(-100, -100)})})
mad = D(1, "a", {"I": T(0, 0), "E": D(2, "b", {"B": T(10, -10), "N": war()})})
labm = lambda c: (c[0]["a"] + c[0]["w1"], c[1]["b"] + c[1]["w2"])
assert sorted(map(labm, pure_nash(mad))) == sorted(
    [("IR", "Nr"), ("IR", "Nd"), ("ID", "Nr"), ("ID", "Nd"), ("ED", "Br"), ("ED",
 ↪  "Bd")])  # p. 164
assert sorted(map(labm, spe(mad))) == [("ED", "Bd"), ("IR", "Nr")]
 ↪  # p. 166


# --- Cabral's entry game and the value of a bond, Figs. 7.7-7.8 (pp. 168-170)
 ↪  -----------
def cabral(M=50, DE=10, DI=20, FE=-10, FI=10, bond=40):
    """Entrant (player 1) then incumbent (player 2). Payoffs (entrant, incumbent).
    Without the bond, accommodation after entry pays the incumbent DI; with it, DI -
 ↪  bond."""
    entry = lambda di: D(1, None, {"out": T(0, M), "in": D(2, None, {"acc": T(DE,
 ↪  di), "fight": T(FE, FI)})})
    return entry(DI), entry(DI - bond)


def name(tree, tag):
    """Give each information set a unique name (cabral() builds them anonymously)."""
    if tree[0] == "T":
        return tree
    _, i, _, moves = tree
    return D(i, f"{tag}{i}", {a: name(c, tag) for a, c in moves.items()})


no_bond, with_bond = (name(t, k) for t, k in zip(cabral(), ("n", "b")))
assert sorted(map(label, pure_nash(no_bond))) == [("in", "acc"), ("out", "fight")]
 ↪  # p. 168
assert list(map(label, spe(no_bond))) == [("in", "acc")]
 ↪  # p. 169
v0, _ = backward_induction(no_bond)
v1, _ = backward_induction(with_bond)
assert v0 == (10, 20) and v1 == (0, 50) and v1[1] - v0[1] == 30   # value of
 ↪  commitment, p. 170
assert backward_induction(name(cabral(FI=-10)[1], "t"))[0] == (0, 50)   # p. 169
 ↪  text's -10: same answer
```

```
# --- MWG Example 9.B.1 (pp. 268-269): Out (0,2); In-Fight (-3,-1); In-Accommodate
 ↪  (2,1) --
mwg = D(1, "E", {"out": T(0, 2), "in": D(2, "I", {"fight": T(-3, -1), "acc": T(2,
 ↪  1)})})
assert sorted(map(label, pure_nash(mwg))) == [("in", "acc"), ("out", "fight")]
assert list(map(label, spe(mwg))) == [("in", "acc")]
```

The last assert of the Cabral block is the source-detail check of L03-06-credibility-entry-commitment §1: with the text's −10 instead of the figure's 10 the answer is unchanged.

---

2 · Mixed versus behavioral, and the absent-minded driver → L03-02-strategies

```
# --- Mixed vs behavioral strategies, Fig. 7.9 (pp. 140-142)
 ↪  ------------------------------
def behavioral_from_mixed(p):
    """Player 2's mixed strategy (p_oo, p_of, p_fo, p_ff) -> (Pr{o|O}, Pr{o|F})."""
    p_oo, p_of, p_fo, p_ff = p
    return p_oo + p_of, p_oo + p_fo


for p in [(1/3, 0, 1/6, 1/2), (1/6, 1/6, 1/3, 1/3)]:            # the book's two
 ↪  mixes, p. 142
    assert isclose(sum(p), 1) and all(isclose(a, b) for a, b in
     ↪  zip(behavioral_from_mixed(p), (1/3, 1/2)))


# --- Absent-minded driver, Fig. 7.10 (p. 142)
 ↪  ------------------------------------------
driver = lambda p: 0 * p + 4 * p * (1 - p) + 1 * (1 - p) ** 2       # = -3p^2 + 2p +
 ↪  1
grid = [k / 3000 for k in range(3001)]
p_star = max(grid, key=driver)
assert isclose(p_star, 1 / 3) and isclose(driver(1 / 3), 4 / 3)
```

---

3 · Stackelberg and Cournot → L03-07-stackelberg

```
from math import isclose, log


def br(q_other, a, b, c):
    """Best response to an observed (or believed) rival quantity, P = a - bQ, cost c:
    max(0, (a - c - b q_other) / (2b))  -- eq. (8.1) with a=100, b=1, c=10."""
    return max(0.0, (a - c - b * q_other) / (2 * b))


def profits(q1, q2, a, b, c1, c2):
    P = max(0.0, a - b * (q1 + q2))
    return P, (P - c1) * q1, (P - c2) * q2
```

```
def cournot(a, b, c1, c2):
    """Simultaneous-move Nash equilibrium, corners included.
    Interior: q_i = (a - 2c_i + c_j) / (3b)."""
    q1, q2 = (a - 2 * c1 + c2) / (3 * b), (a - 2 * c2 + c1) / (3 * b)
    if q1 < 0:
        q1, q2 = 0.0, br(0, a, b, c2)
    elif q2 < 0:
        q1, q2 = br(0, a, b, c1), 0.0
    return (q1, q2) + profits(q1, q2, a, b, c1, c2)


def stackelberg(a, b, c1, c2):
    """Subgame-perfect outcome with firm 1 leading. The leader maximizes
    pi_1(q1, R2(q1)) where R2 = br(., a, b, c2) is kinked at q1 = (a - c2)/b; each
 ↪  piece
    is concave, so compare the best point on each piece."""
    kink = max(0.0, (a - c2) / b)
    qa = min(max((a - 2 * c1 + c2) / (2 * b), 0.0), kink)     # follower active
    qb = max((a - c1) / (2 * b), kink)                          # follower shut out
    pi = lambda q: profits(q, br(q, a, b, c2), a, b, c1, c2)[1]
    q1 = max((qa, qb), key=pi)
    q2 = br(q1, a, b, c2)
    return (q1, q2) + profits(q1, q2, a, b, c1, c2)


# --- Tadelis Section 8.3.2 (pp. 160-161): p = 100 - q, c = 10
 ↪  ----------------------
S, C = stackelberg(100, 1, 10, 10), cournot(100, 1, 10, 10)
assert S[:2] == (45.0, 22.5) and isclose(S[3], 1012.5) and isclose(S[4], 506.25)
assert C[:2] == (30.0, 30.0) and isclose(C[3], 900) and isclose(C[4], 900)
assert isclose(S[3] - C[3], 112.5) and isclose(S[4] - C[4], -393.75)   # first-mover
 ↪  advantage
assert isclose(S[2], 32.5) and isclose(C[2], 40)                       # price falls

# --- Gibbons Section 2.1.B (pp. 62-63): general (a, c) ----------------------------
for a, c in [(100, 10), (50, 5), (12, 3)]:
    S, C = stackelberg(a, 1, c, c), cournot(a, 1, c, c)
    assert isclose(S[0], (a - c) / 2) and isclose(S[1], (a - c) / 4)
    assert isclose(S[0] + S[1], 3 * (a - c) / 4) and isclose(C[0] + C[1], 2 * (a - c)
     ↪  / 3)
    assert S[3] > C[3] and S[4] < C[4] and S[3] + S[4] < C[3] + C[4]

# --- Unobserved q1 (Gibbons p. 63): if 2 expects (a-c)/2, 1 would rather play
 ↪  3(a-c)/8 --
a, c = 100, 10
assert isclose(br(br((a - c) / 2, a, 1, c), a, 1, c), 3 * (a - c) / 8)
```

```
# --- Nash but not subgame perfect (pp. 162-163) ----------------------------------------
def leader_profit(q1, strategy2, a=100, c=10):
    return profits(q1, strategy2(q1), a, 1, c, c)[1]


flat30 = lambda q1: 30.0                                        # q2(q1) = 30 for every q1
grid = [k / 4 for k in range(0, 401)]
assert max(grid, key=lambda q: leader_profit(q, flat30)) == 30.0   # q1 = 30 is a BR to it
assert br(30, 100, 1, 10) == 30.0                               # and 30 is a BR on the path
assert br(45, 100, 1, 10) != flat30(45)                         # but off the path it is not


def flood(q_hat):
    """Tadelis's remark: best-respond to q_hat, flood the market (q2 = 100) otherwise."""
    return lambda q1: br(q_hat, 100, 1, 10) if q1 == q_hat else 100.0


assert br(20, 100, 1, 10) == 35.0     # the remark's on-path quantity is 35 (p. 163 prints 40)
for q_hat in [5.0, 20.0, 45.0, 60.0, 85.0]:                     # a continuum of NE outcomes
    s2 = flood(q_hat)
    assert max(grid + [q_hat], key=lambda q: leader_profit(q, s2)) == q_hat
    assert s2(q_hat) == br(q_hat, 100, 1, 10)
```

`stackelberg` is exact: the leader's reduced profit is concave on each side of the kink $q_1 = (a - c_2)/b$ where the follower exits, so the optimum is the better of the two pieces' optima. The companion's JavaScript uses the same expression and companions/check_companions.py verifies it against a brute-force grid search.

---

4 · Hyperbolic discounting → L03-08-time-inconsistency

```
# --- Section 8.3.4 (pp. 166-169): beta = 1/2, delta = 1, u = ln ------------------------
def v1(x1, K=1.0, beta=0.5):
    """Self 1's payoff when self 2 best-responds with x2 = 2K2/3, x3 = K2/3 (p. 168)."""
    K2 = K - x1
    return log(x1) + beta * log(2 * K2 / 3) + beta * log(K2 / 3)


xs = [k / 20000 for k in range(1, 20000)]
x1_star = max(xs, key=v1)
```

```
assert isclose(x1_star, 0.5)                          # the correct derivative gives x1
 ↪  = K/2
assert v1(0.5) > v1(2 / 3)                            # p. 169 prints 2K/3: not the
 ↪  maximum
# self 1's own preferred plan (commitment): ln x1 + .5 ln x2 + .5 ln x3 -> (1/2, 1/4,
 ↪  1/4)
best = max(((x1, x2) for x1 in xs[::40] for x2 in xs[::40] if x1 + x2 < 1),
           key=lambda z: log(z[0]) + 0.5 * log(z[1]) + 0.5 * log(1 - z[0] - z[1]))
assert all(isclose(u, w, abs_tol=2e-3) for u, w in zip(best, (0.5, 0.25)))
```

The second assert is the correction of L03-08-time-inconsistency §2.2: the book's printed $x_1 = \frac{2}{3}K$ is not the maximum of the objective it prints.

---

5 · Run everything

```
python Notas/L03/figures/make_figures.py          # 7 figures + their asserts
python Notas/L03/companions/check_companions.py   # both companions vs Python, needs
 ↪  node
```